Certified symmetry discovery for polynomial and dictionary vector fields: confidence sets for Lie-algebra generators from noisy vector-field measurements

Ihor Kendiukhov[1*]

[1*]Institute of Medical Genetics and Applied Genomics, University of Tübingen, Tübingen, Germany.

Corresponding author(s). E-mail(s): ihor.kendiukhov@student.uni-tuebingen.de;

Abstract

Continuous symmetries of dynamical systems are often discovered from data as the nullspace of a linear operator. With noisy data, a symmetry is declared whenever a singular value looks small. This has no error control and points the wrong way: failing to reject a symmetry is not evidence for it. We recast symmetry discovery as an equivalence test: a generator is certified as an approximate symmetry only when the data rule out a violation above a stated tolerance. For vector-field models linear in their coefficients, such as polynomial or dictionary models, a relative measure of symmetry violation is a ratio of norms of linear images of the coefficients, so one confidence region yields bounds for every candidate generator at once. On ten systems with exactly known symmetries and 72,000 simulated data sets, the procedure never certified a false symmetry, whereas a fixed threshold at the same tolerance did so in 0.5% of runs on average and 63% in its worst setting. The price is fewer detections (48.5% against 78.4%). The guarantee requires a model class containing the true system, exactly observed states and a confidence region matched to the error structure; violating the first two can manufacture false symmetries.

Keywords: Symmetry discovery, Lie algebra, Equivariance, Equivalence testing, Confidence sets, Dynamical systems

1 Introduction

Symmetry is one of the few inductive biases that reliably pays for itself in machine learning, from group-equivariant convolutions [1] to general constructions for arbitrary matrix groups [2] and the geometric programme that organises them [3]. All of it presumes that the symmetry is known. When it is not, it has to be discovered from data, and the question this paper is about is what a discovery may honestly claim.

A continuous symmetry of a dynamical system $\dot{x} = F(x)$ is a one-parameter family of state transformations that leaves the dynamics unchanged. Each such family is generated by a vector field ξ, and ξ generates a symmetry exactly when the Lie bracket $[\xi, F]$ vanishes identically. Because the bracket is linear in ξ, the set of generators is the nullspace of a linear operator. This observation drives a large body of work: it is how symmetries of point clouds [4], of trained networks [5, 6] and of governing equations [7] are found, and it was recently unified and substantially generalised by Otto et al. [8], who show that enforcing, discovering and promoting symmetry are all governed by the same Lie-derivative operator, with discovery reduced to computing a nullspace.

With real data the operator must be estimated, and an estimated operator has no exact nullspace. Practice therefore replaces "is this singular value zero?" with "is this singular value small?", by comparing it to a tolerance or by looking for a gap in the spectrum. This fails in two ways.

The tolerance means nothing in particular. A singular value of the Lie-derivative operator has the units of the data and depends on the arbitrary scaling of the basis chosen for the candidate generators; a tolerance of 10^{-3} is a statement about neither the system nor the evidence. Whether a small value indicates a symmetry or merely a fortunate draw of the noise depends on the noise level, the sample size, the conditioning of the design and the dimension of the model class, none of which the tolerance sees.

The burden of proof is inverted. Suppose we improve matters by attaching a hypothesis test, as a rank test would [9, 10]: test H_0: "this generator is an exact symmetry" and declare a symmetry when the test does not reject. Such a test is valid, rejecting a true symmetry at most 5% of the time, and it is still the wrong instrument, because *failing to reject H_0 is not evidence for H_0*. The error we care about, declaring a symmetry that does not exist, is the type-II error, and no level-α test controls it. In our experiments this is not a technicality: a valid level-0.05 rank test declares a symmetry that is not one in 7% of replications on average over our settings, and in every replication of its worst setting.

Our reframing.

A procedure that may honestly say "ξ is a symmetry" must be able to *rule out* the alternative. So we ask for an *equivalence test* [11–13]: fix a tolerance δ, and certify ξ only when the data reject the hypothesis that its defect exceeds δ. This buys error control in the direction that matters: the probability of certifying a generator whose true defect exceeds δ is at most α. It also forces the tolerance into the open, where it can be argued about.

Two ingredients make it practical. First, we measure symmetry violation with a *relative* defect: the norm of the bracket $[\xi, F]$ divided by the norm of the two terms whose cancellation produces it (Definition 1). The resulting number lies in $[0, \sqrt{2}]$, does not change when ξ or F is rescaled or the candidate class is reparameterised, and answers a concrete question: by what fraction do the two halves of the bracket fail to cancel? Once the region of interest and the scales of the state variables are declared, a tolerance in these units means the same thing for every system. Second, for a model that is linear in its coefficients β (a polynomial vector field, or any fixed dictionary of functions), the squared defect is a *ratio of squared norms of linear images of* β (Lemma 2). A confidence ellipsoid for β therefore propagates through it by optimising over an ellipsoid, which reduces to trust-region subproblems [14, 15] that can be solved globally and with certified accuracy. Because a single event (β lies in its confidence region) implies the bound for *every* generator at once, the certificate is simultaneous over the whole candidate space, with no correction for selection or multiplicity and no sample splitting. Each of these steps is elementary on its own; the contribution is the formulation that makes them fit together.

What we find.

We evaluate on ten polynomial dynamical systems whose symmetry algebras we compute exactly by rational linear algebra, so that every error is scored against ground truth rather than against a tolerance. Across 240 settings of system, sample size and noise level (72,000 replications), our procedure, SymCert, never once certifies a generator whose true defect exceeds the tolerance (the one-sided 95% upper bound on its error rate is 0.010 in every setting). A fixed threshold at the same tolerance errs in 0.5% of replications on average and in 63% in its worst setting; an eigengap heuristic and the rank and split-sample significance tests reach 100%, 100% and 92% in their worst settings. This error control costs detection power, and we report the cost. Two non-polynomial systems fitted with dictionaries that contain them behave in the same way. On real ocean drifter velocities, the procedure certifies the rotation invariance that the physics of near-inertial currents predicts, to a relative defect of 0.074, under assumptions stated with the result.

Four further findings, each a consequence of taking the statistics seriously.

Additive derivative noise rarely made a system without symmetry look symmetric in our tests, but it can tilt a real symmetry. When the right-hand side is measured with noise, the class is correctly specified and the states are exact, the plug-in defect of a direction near a true symmetry is biased *upward*, which costs detections. On a system without symmetry the smallest plug-in defect, selected for being small, can fall below its true value, and a threshold at the tolerance then declared a false symmetry in at most 3.7% of replications, and only when the smallest true defect was within a factor of about two of the tolerance, never at four times it. The larger failure is on symmetric systems: at high noise the estimated subspace tilts, and a fixed threshold that declares the right number of generators can then declare some whose true defect exceeds the tolerance; this is the source of the threshold rule's 63% worst case, on the planar rotation at the highest noise level. SymCert certified no false symmetry in any of these tests.

Coverage has a computable price. If the data occupy a region μ while the claim is about a domain ν, the defect on ν can exceed the defect measured on μ by at most an explicit factor built from two generalised eigenvalues of Gram matrices (Proposition 6); the factor is infinite exactly when the data lie in the zero set of a nonzero field of the finite-dimensional space that contains every bracket, as happens, to numerical precision, for a single trajectory that converges to an algebraic limit cycle such as a circle; single trajectories of the Van der Pol and Lorenz systems give large but finite factors. Across 50 coverage settings, the nullspace computed on the data declared spurious symmetry only beyond the threshold this factor sets, and in none of the 13 settings below it, where the proposition rules out, at population level, declaring more generators than the true algebra has; SymCert declared none in any setting.

The model class and noisy states can manufacture symmetry. Fitting a class too small to contain the truth produces a model with symmetries the system does not have; sparse regression, the standard tool of data-driven model discovery [16], does this automatically on a weakly symmetry-breaking system by deleting the term that breaks the symmetry; and noise in the states makes least squares estimate a pseudo-true field, which on a hardening Duffing oscillator was more symmetric than the system, enough for the certificate itself to certify a false symmetry, while on a symmetric Hopf system it hid the symmetry instead. Our certificate does not repair these failures on its own, because its validity is conditional on a correctly specified class and exactly observed states. A lack-of-fit test is not a remedy either: we quantify how large a symmetry violation such a test of given power can miss (Proposition 3), and give a procedure that remains valid when the class is chosen from the data, provided the largest class considered contains the truth (Proposition 4).

Units matter unless the metric is declared. With a Euclidean norm on the state, changing the units of one coordinate can turn a clear absence of symmetry into an apparent one; declaring the scales of the state variables removes the dependence (Lemma 1).

Contributions.

- A relative equivariance defect (Definition 1) that is invariant to the arbitrary choices in the nullspace formulation and, when the output metric is transported with the coordinates, to every affine change of state coordinates (Lemma 1); minimising it is a generalised eigenvalue problem.
- SymCert: upper and lower confidence bounds on the defect of the *true* system, finite-sample valid under Gaussian homoskedastic errors and as valid as the supplied coefficient region otherwise, simultaneously over all candidate generators and all tolerances, and hence over any data-dependent selection rule (Proposition 1). The bounds are the supremum and infimum of the defect over the confidence ellipsoid, not a ratio of separate bounds on its numerator and denominator, and are evaluated with certified numerics whose error can only widen them (Proposition 2). From the same region the procedure certifies approximate symmetry, rules out every approximate symmetry below a stated tolerance in the candidate class, and reports a certified plug-in subspace of approximate generators.

- An explicit account of what the guarantee covers (Table 1), including how large a symmetry violation a lack-of-fit test of given power can miss (Proposition 3) and certificates that stay valid when the model class is chosen from the data within a nested family whose largest member contains the truth (Proposition 4).
- A non-asymptotic Le Cam lower bound (Proposition 5) on the tolerance any procedure with false-certification rate α can certify; at a fixed model class, SymCert matches its $\sigma/\sqrt{N}$ scaling with a constant gap that we report, together with its growth in the number of coefficients.
- A quantitative account of the mechanisms that create apparently valid symmetries: selection optimism, incomplete coverage, model truncation, sparse regression, noisy states and units, together with evidence that, in the systems tested, additive derivative noise made a system without symmetry look symmetric only rarely (in at most 3.7% of replications) and only when its smallest true defect was within about a factor of two of the tolerance, although in a symmetric system it can tilt the subspace that a threshold declares.
- An open-source implementation, exact ground truth for every polynomial benchmark, a full record of the compute, and a test suite.

2 Results

2.1 Symmetry discovery as a nullspace problem

Generators and the Lie derivative.

Let $\dot{x} = F(x)$ be a dynamical system on $\mathbb{R}^n$ and let ξ be a vector field on $\mathbb{R}^n$. The flow of ξ is a one-parameter group of state transformations, and it maps solutions of $\dot{x} = F(x)$ to solutions if and only if the Lie bracket

$$\mathcal{L}_\xi F \;=\; [\xi, F] \;=\; DF\,\xi \;-\; D\xi\,F \tag{1}$$

vanishes identically [17, Ch. 2]; here DF and $D\xi$ are Jacobian matrices. Equation (1) is the Lie derivative in the sense of Otto et al. [8, Eq. 16]; the same expression, with $D\xi\,F$ replaced by the appropriate output action, governs symmetries of a learned map, and with $D\xi\,F$ deleted it governs invariances of a learned scalar function (Supplementary Note 1).

We search inside a *candidate class* $\mathcal{V} = \mathrm{span}\{\zeta_1, \ldots, \zeta_P\}$ of vector fields, writing $\xi_\theta = \sum_p \theta_p \zeta_p$; in our experiments $\mathcal{V}$ is usually the set of affine generators $\xi(x) = Sx + b$, whose flows form the affine group, or all polynomial fields of degree at most two; the scaling study and the invariance study of Supplementary Note 1 use the linear generators $\xi(x) = Sx$. Because equation (1) is linear in ξ, the exact symmetry algebra inside $\mathcal{V}$ is the nullspace $\{\theta : [\xi_\theta, F] \equiv 0\}$, and it is standard to obtain it as the nullspace of the positive semi-definite Gram operator $\mathcal{S}$, defined by $\langle \xi', \mathcal{S}\xi \rangle = \int (\mathcal{L}_{\xi'} F)^\top (\mathcal{L}_\xi F)\, d\mu$, where μ is the distribution of the observed states [8, §7].

The target domain and the output metric.

A symmetry is a statement about a region of state space and about how vector fields are measured there. We therefore fix a *target measure* ν, the region over which the user wants the symmetry to hold, and a positive-definite *output metric* W, and measure vector fields by $\|v\|^2_{\nu,W} = \int v(x)^\top W v(x)\, d\nu(x)$. In most of our experiments ν is uniform on a box, whose monomial moments are available in closed form, and in the others the integrals are closed-form moments or quadratures that are exact or accurate to rounding, so *the geometry used by the certificates carries no Monte Carlo error*; all statistical uncertainty lives in the model. (Monte Carlo integration is used only to evaluate reference quantities of the known truth, such as the true defects of non-polynomial systems.) Unless stated otherwise the state variables share a scale and $W = I$. Keeping ν separate from μ matters: their mismatch is the incomplete-coverage problem studied below.

The defect.

Two quantities cancel in equation (1). Measuring the residual relative to their size gives a dimensionless number.

Definition 1 (Relative equivariance defect)**.** For a generator ξ and a field F,

$$\rho_\nu(\xi; F) \;=\; \frac{\big\| DF\,\xi - D\xi\,F \big\|_{\nu,W}}{\big(\|DF\,\xi\|^2_{\nu,W} + \|D\xi\,F\|^2_{\nu,W} \big)^{1/2}}, \tag{2}$$

defined whenever the denominator is positive. When it vanishes, that is, when $DF\,\xi$ and $D\xi\,F$ are both zero ν-almost everywhere, the defect is undefined, and such a generator is never certified.

Lemma 1 (Properties)**.** *Where it is defined, $\rho_\nu \in [0, \sqrt{2}]$, and $\rho_\nu(\xi; F) = 0$ if and only if $[\xi, F] = 0$ ν-almost everywhere, that is, on the support of ν for continuous fields. When that support is the closure of an open set, as for every target measure used here, this holds if and only if ξ generates a symmetry of F on its interior, and for polynomial (or other analytic) ξ and F the bracket then vanishes identically. On a measure concentrated on a lower-dimensional set, such as data on a limit cycle that lies on an algebraic curve, a zero defect need not indicate any symmetry (Proposition 6). The defect is invariant under $\xi \mapsto c\,\xi$ and $F \mapsto c'F$ for nonzero c, c', hence under any invertible reparameterisation of the candidate class. Under an invertible affine change of coordinates $x = Az + b$, with ν transported to the new coordinates and W replaced by $A^\top W A$, it is unchanged. With the metric held fixed at $W = I$, invariance holds in general only for similarity transformations ($A = cO$, O orthogonal, $c > 0$).*

The proof is in Methods. ρ is the fraction of the bracket's two terms that fails to cancel, so $\rho = 0.05$ means the two halves of the bracket agree to 5% in root-mean-square over the target domain. A generator with $\rho_\nu(\xi; F) \le \delta$ is what we call a *δ-approximate symmetry*; δ plays the role of the singular-value tolerance of the

Gram-operator approach, but it does not depend on the scale of F, on the scale of ξ, or on how the candidate basis is chosen.

It does depend on the units of the state when W is held fixed. Measuring one coordinate of a Lotka–Volterra system in units a hundred times smaller lowers the smallest Euclidean defect in the affine class from 0.27 to 0.0035 (Supplementary Table S1), which would turn a clear absence of symmetry into an apparent one. The remedy is to declare the metric: either nondimensionalise each coordinate by a characteristic scale s_i of the problem, or, equivalently, use $W = \mathrm{diag}(1/s_i^2)$ with s_i expressed in the current units. Either way the defect, and hence the meaning of a tolerance, becomes independent of the units.

One class of generators is outside the reach of the defect. A translation $\xi = b$ has $D\xi = 0$, so the second term of the bracket vanishes and $\rho_\nu(b; F)$ is 1 whenever $DF\, b \neq 0$ and undefined when it is zero: the ratio says nothing about how close a translation is to a symmetry. Certifying translation symmetries needs a different normalisation of $\|DF\, b\|$; the machinery below applies to any normalisation that is the norm of a linear image of the model coefficients, but we do not pursue this here.

Remark 1 (Small eigenvalues are hard to read)**.** Even with a scale fixed, the smallest eigenvalue of an estimated Gram matrix is a delicate object: its fluctuations are of a different order from those of the bulk, and the classical asymptotics [18] and their random-matrix refinements [19] show that the map from "small" to "zero in the population" is not one a fixed threshold can implement. This is why the certificate below is stated for the defect ratio directly rather than for eigenvalues.

Remark 2 (An average, not a supremum)**.** ρ_ν is a mean-square quantity, so a generator can have a small defect while violating the symmetry badly on a small part of the domain, a property it shares with the Gram-operator formulation it refines and with every L^2-based criterion. Two responses are available, each with a cost: choose ν to weight the region whose behaviour matters, which is the intended use of making ν explicit; or replace the numerator by a supremum norm, which for polynomials is bounded by a constant times the L^2 norm through a Markov-type inequality, at the cost of that constant.

Without a normalisation the "small singular value" criterion is not even well posed: rescaling one basis generator ζ_p by 10^3 multiplies the off-diagonal entries of the p-th row and column of the Gram matrix by 10^3 and its (p, p) entry by 10^6, moving eigenvalues across any fixed threshold. ρ is a functional of the vector field ξ itself and is immune to this.

Model and data.

We take the measured-derivative design of data-driven model discovery [16]: pairs $(x_i, y_i)_{i=1}^N$ with $x_i \sim \mu$ and

$$y_i \;=\; F(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2 I_n) \text{ i.i.d.}, \tag{3}$$

and a model class $F_\beta(x) = \sum_{i,a} \beta_{i,a}\, \phi_a(x)\, e_i$ that is linear in its coefficients $\beta \in \mathbb{R}^Q$. In most of the paper the dictionary $\{\phi_a\}$ consists of the monomials of degree at most d, so that $Q = n\binom{n+d}{d}$; any fixed dictionary of differentiable functions can be used instead (trigonometric or radial libraries, for example), and we use two such libraries for non-polynomial systems. Ordinary least squares gives $\hat\beta$ and, in this homoskedastic Gaussian model, an *exact* finite-sample confidence ellipsoid. Throughout, σ is reported relative to the root-mean-square size of F over the target domain, so "noise level 0.1" means the noise standard deviation is a tenth of the typical size of the right-hand side.

Lemma 2 (Bilinear structure)**.** *For a model that is linear in a fixed dictionary and a finite-dimensional candidate class $\mathcal{V}$, there are matrices $L_\theta, M_\theta^{(1)}, M_\theta^{(2)}$, linear in θ and independent of the data, such that*

$$\rho_\nu(\xi_\theta; F_\beta)^2 \;=\; \frac{\|L_\theta \beta\|^2}{\|M_\theta^{(1)}\beta\|^2 + \|M_\theta^{(2)}\beta\|^2} \;=\; \frac{\|L_\theta\beta\|^2}{\|M_\theta \beta\|^2}, \qquad L_\theta = M_\theta^{(1)} - M_\theta^{(2)}, \tag{4}$$

where $\|\cdot\|$ is the Euclidean norm and M_θ stacks $M_\theta^{(1)}$ and $M_\theta^{(2)}$. Consequently, over the generators whose defect is defined, $\min_\theta \rho_\nu(\xi_\theta; F_\beta)^2$ is the smallest generalised eigenvalue of the pencil $(C(\beta), D(\beta))$ with $\theta^\top C\theta = \|L_\theta\beta\|^2$ and $\theta^\top D\theta = \|M_\theta\beta\|^2$, restricted to the range of $D(\beta)$; the null space of $D(\beta)$ consists exactly of the generators with an undefined defect.

The matrices are obtained by expanding equation (1) in the dictionary and absorbing the integrals over ν (and the metric W) into a symmetric square root; for monomials and a box, ball, annulus or Gaussian target the integrals are closed-form moments, and for other dictionaries they are computed by quadrature rules that are exact, or accurate to rounding, for the integrands involved (Methods). The reduction to a generalised eigenproblem makes the plug-in symmetry directions independent of how the candidate basis is scaled.

2.2 SymCert: certificates for the defect

The plug-in defect $\rho_\nu(\xi_\theta; F_{\hat\beta})$ is an estimate. What a practitioner needs is a statement about the *true* system: an upper bound on $\rho_\nu(\xi_\theta; F_{\beta^\star})$ that holds with a stated probability, for whichever generator the analysis happens to select.

The simultaneous certificate.

Let $\mathcal{B} \subseteq \mathbb{R}^Q$ be a confidence region for the true coefficients $\beta^\star$ with $\mathbb{P}(\beta^\star \in \mathcal{B}) \geq 1 - \alpha$, constructed in any way. For every $\theta \neq 0$ define

$$U(\theta) \;=\; \sup_{\beta\in\mathcal{B}} \rho_\nu(\xi_\theta; F_\beta), \qquad L(\theta) \;=\; \inf_{\beta\in\mathcal{B}} \rho_\nu(\xi_\theta; F_\beta), \tag{5}$$

with the convention that $U(\theta) = +\infty$ if the denominator of the defect vanishes for some $\beta \in \mathcal{B}$ (the generator cannot then be certified), and that the infimum is taken

over the β for which the defect is defined. The lower bound $L(\theta)$ is not to be confused with the matrix L_θ of Lemma 2.

Proposition 1 (Simultaneous validity)**.** *If* $\mathbb{P}(\beta^\star \in \mathcal{B}) \geq 1-\alpha$*, then* $\mathbb{P}\big(L(\theta) \leq \rho_\nu(\xi_\theta; F_{\beta^\star}) \leq U(\theta) \text{ for every } \theta \text{ with a defined defect}\big) \geq 1-\alpha$*. Consequently, for* any *rule, deterministic or data-dependent, that selects generators and certifies those with* $U \leq \delta$*, and for any tolerance, fixed in advance or not,*

$$\mathbb{P}\big(\exists\, \theta, \delta:\ U(\theta) \leq \delta < \rho_\nu(\xi_\theta; F_{\beta^\star})\big) \ \leq\ \alpha. \tag{6}$$

The proof is one line (Methods): the single event $\{\beta^\star \in \mathcal{B}\}$ implies the bound for every generator and every tolerance at once, so the family-wise false-certification guarantee (6) comes for free. No multiplicity correction is applied because none is needed, and no data splitting is required to protect against the selection of θ, the problem that motivates split-sample and post-selection inference [20–23]. The guarantee is exactly as good as the region $\mathcal{B}$: it inherits $\mathcal{B}$'s coverage and nothing more, and it says nothing about choices made outside the candidate class, such as the model class, the dictionary or the target domain (Section 2.3).

Computation.

Take $\mathcal{B}$ to be an ellipsoid $\{\hat\beta + \hat\Sigma^{1/2} u : \|u\| \leq R\}$.

Proposition 2 (Evaluation)**.** *(i) Each of* $\sup_{\mathcal{B}} \|L_\theta \beta\|$ *and* $\inf_{\mathcal{B}} \|M_\theta \beta\|$ *is the extremum of* $\|a + Bu\|$ *over the ball* $\|u\| \leq R$*, a trust-region subproblem, so the* separate bound $U_{\text{sep}}(\theta) = \sup_{\mathcal{B}} \|L_\theta\beta\| / \inf_{\mathcal{B}} \|M_\theta\beta\| \geq U(\theta)$ *costs two of them (with* $U_{\text{sep}}(\theta) = +\infty$ *when* $\inf_{\mathcal{B}} \|M_\theta\beta\| = 0$*). (ii) If* $\inf_{\mathcal{B}} \|M_\theta\beta\| > 0$ *(otherwise* $U(\theta) = +\infty$ *by the convention of (5)), then for every level* $t \geq 0$*,* $U(\theta)^2 \leq t$ *if and only if* $\min_{\|u\|\leq R}\ t\,\|M_\theta\beta(u)\|^2 - \|L_\theta\beta(u)\|^2 \geq 0$*, again a trust-region subproblem, now with an indefinite quadratic; such problems are solved globally because their Lagrangian dual has no duality gap [15, 24]. A one-dimensional search on* t *therefore yields* $U(\theta)$*, and likewise the lower bound* $L(\theta)$*, to any prescribed relative tolerance. (iii) Under model (3) with ordinary least squares and an* $N \times (Q/n)$ *design matrix* Φ *of the dictionary of full column rank, taking* $\hat\Sigma$ *the usual covariance estimate and* $R^2 = Q\,\mathrm{F}_{Q,\,\mathrm{dof};\,1-\alpha}$*, where* $\mathrm{F}_{Q,\,\mathrm{dof};\,1-\alpha}$ *is a quantile of the* F *distribution and* $\mathrm{dof} = nN - Q$*, makes* $\mathcal{B}$ *an* exact $1-\alpha$ *confidence region in finite samples, conditional on the design.*

Every quantity entering U and L is evaluated through a Lagrangian dual bound rather than at the point where a root finder stops, so the computed U can only err upwards and the computed L only downwards; the root-finding tolerance (a relative 10^{-7} on $U(\theta)$ and $L(\theta)$ themselves, with a further relative safety margin of 10^{-9} for rounding) affects tightness, never validity (Methods). The bounds are therefore *finite-sample valid* whenever $\mathcal{B}$ is an exact region, and the word *exact* is reserved here for the coverage of $\mathcal{B}$ and for the trust-region extrema themselves. The test suite checks the solvers against brute-force search over the ball.

Optimising the ratio directly, rather than its numerator and denominator separately, matters little where the bounds are informative: across the 168 settings of the resolution study the sharp bound is narrower than U_{sep} by a median of 2.5%, and by up to a third only where both bounds are large ($N = 100$, noise level 0.3). The dominant source of conservatism is not the separation. The gap between U and the lower bound of Proposition 5 grows with the number Q of model coefficients, but nearly as much for the pointwise bound below, so it is not mainly a cost of simultaneity (Section 2.9).

A tighter bound for a pre-specified generator.

When θ is fixed in advance, or chosen on a held-out split, the ellipsoid can be replaced by Gaussian concentration of $\|L_\theta(\hat{\beta} - \beta^\star)\|$ about its mean, giving a width driven by an *effective* dimension $\mathrm{tr}(\Gamma_\theta)/\|\Gamma_\theta\|_{\mathrm{op}}$ rather than by the full parameter count. Here $\Gamma_\theta = c\, L_\theta \hat{\Sigma} L_\theta^\top$, with $c = \mathrm{dof}/\chi^2_{\mathrm{dof};\,\alpha_\sigma}$ a chi-square upper confidence factor for the noise variance. Concretely, $\|\Gamma_\theta^{1/2} z\|$ with z standard Gaussian has mean at most $(\mathrm{tr}\,\Gamma_\theta)^{1/2}$ and is a Lipschitz function of z with constant $\|\Gamma_\theta\|_{\mathrm{op}}^{1/2}$, so the Gaussian concentration inequality [25–27] bounds $\|L_\theta(\hat{\beta} - \beta^\star)\|$ by $w_L = (\mathrm{tr}\,\Gamma_\theta)^{1/2} + (2\|\Gamma_\theta\|_{\mathrm{op}} \log(1/\alpha_L))^{1/2}$ with probability at least $1 - \alpha_L - \alpha_\sigma$. The denominator is bounded in the same way (w_M, with M_θ in place of L_θ, at level α_M), and the upper bound is $(\|L_\theta\hat{\beta}\| + w_L)/(\|M_\theta\hat{\beta}\| - w_M)$, or $+\infty$ if $w_M \geq \|M_\theta\hat{\beta}\|$. With $\alpha_L = 0.98\alpha$ and $\alpha_M = \alpha_\sigma = 0.01\alpha$ its coverage is at least $1 - \alpha$ (Methods). We report it for comparison.

Certify.

ξ_θ is certified as a δ-approximate symmetry when $U(\theta) \leq \delta$. Equivalently, the smallest tolerance the data support for that generator, which we call its certified tolerance, is $U(\theta)$ itself.

Refute.

If $\min_\theta L(\theta) > \delta$ then, with confidence $1 - \alpha$, the system has *no* δ-approximate symmetry in the candidate class. A lower bound on this minimum follows from the same region via smallest and largest singular values (Methods).

The certified plug-in subspace.

Let V_r be spanned by the r generalised eigenvectors of $(C(\hat{\beta}), D(\hat{\beta}))$ with smallest plug-in defect. By the Courant–Fischer theorem V_r minimises the largest *plug-in* defect among all r-dimensional subspaces of the candidate class. We report the largest r such that every generator in V_r is certified, computed by bounding the largest and smallest singular values of the restricted maps over $\mathcal{B}$, and call it the dimension of the certified plug-in subspace. Two qualifications are part of its meaning. First, it is a lower bound on the largest dimension of any certifiable subspace: the plug-in subspace minimises the plug-in defect, not the upper bound, and the nesting may be suboptimal. Second, it certifies a subspace of approximate generators, not an approximate Lie algebra. For exact symmetries the bracket of two

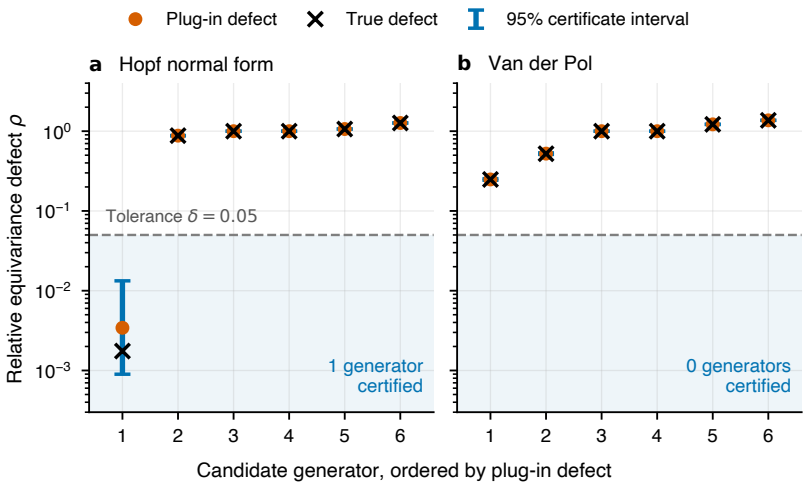


Fig. 1 What the procedure outputs. Two systems at $N = 3200$ and noise level 0.03. Each candidate generator gets an interval for its *true* defect, valid simultaneously across all six. A generator is certified when its interval lies entirely inside the shaded region below the tolerance. On the Hopf normal form the rotation generator is certified; on Van der Pol nothing is, correctly, and the lower certificate goes further, ruling out at 95% confidence the existence of *any* generator in the class with defect below 0.137. For the large defects the intervals are narrower than the plotting symbols: the certificate is only wide where the answer is genuinely uncertain.

symmetries is again a symmetry, but the set of δ-approximate generators need not be a vector space, nor be closed under the bracket. Closure is a separate, checkable property: for generators ξ_a, ξ_b of the subspace we measure the part of $[\xi_a, \xi_b]$ outside the subspace, relative to the size of its two terms, and, when the candidate class is closed under the bracket, as the affine class is, certify the defect of each commutator that is not negligible (Methods, Supplementary Table S2). V_r is data-dependent, which is why Proposition 1 is stated for all θ at once.

Choosing the tolerance after seeing the data.

Because (6) holds uniformly in δ, the whole curve $\delta \mapsto \{\theta : U(\theta) \le \delta\}$ is valid simultaneously, and reporting U for every generator of interest is always safe. A tolerance chosen after inspecting that curve does not invalidate the claim $\rho_\nu \le \delta$; what it gives up is the scientific meaning of a tolerance fixed in advance, so a post hoc δ should be reported as such, together with U.

Figure 1 shows the three read-offs on two systems at a single sample size and noise level: the certified interval for every candidate generator, the tolerance it is compared against, and the resulting dimension of the certified plug-in subspace.

2.3 What the guarantee covers

Proposition 1 is a conditional statement: if the coefficient region covers the true coefficients with probability $1 - \alpha$, the certificate inherits that probability for every generator and every tolerance. Everything else is in the condition. Table 1 lists each

guarantee in the paper with what it requires, and the rest of this subsection takes the requirements in turn.

guarantee	requires	evidence
False-certification rate $\leq \alpha$, simultaneously over generators and tolerances (Proposition 1)	a region with coverage $\geq 1-\alpha$ for coefficients of a class that contains the truth; dictionary, candidate class, target measure ν and metric W fixed without looking at the data	Table 2, Fig. 2a
Exact finite-sample coverage of the region	Gaussian homoskedastic errors on the right-hand side, independent of states that are observed exactly	Table 3
Asymptotic coverage of the region	independent rows (White sandwich), independent trajectories with their number growing (cluster-robust), or long trajectories (Newey–West)	Table 3, Supplementary Fig. S1
Validity when the class is chosen from the data (Proposition 4)	the largest class of a nested family contains the truth; regions of Proposition 2(iii) (Gaussian homoskedastic errors, fixed design, exactly observed states)	Section 2.8
Validity under a misspecified class	an externally justified bound on the omitted part (sensitivity parameter)	Supplementary Fig. S2
Coverage diagnostic κ (Proposition 6)	none to compute it from the observed states; as a bound on the true defect, the fitted class contains the truth (a population-level statement; estimation error not included)	Fig. 2b, Supplementary Table S3
Resolution limit (Proposition 5)	Gaussian errors, fixed design, one generator, fixed model class	Fig. 2c, Supplementary Table S4
Not covered	noisy states (errors in variables); a class chosen by a lack-of-fit test on the same data; translation generators; discrete symmetries	Supplementary Fig. S3, Section 2.8

Table 1 Guarantees and the assumptions they rest on. Each row states what has to hold for the guarantee in the first column; the last column points to the experiments that probe it. The first row is the main result; all others qualify it.

The model class.

Validity needs a coefficient vector $\beta^\star$ with $F = F_{\beta^\star}$, so the fitted class must contain the truth. Three situations have to be kept apart.

(i) A class fixed in advance and correctly specified. This is the setting of Proposition 1, and the false-certification rate is at most α.

(ii) A class selected from the data. A natural workflow enlarges the class until a lack-of-fit F-test against a larger class stops rejecting, and then certifies in the chosen class on the same data. This has no guarantee, for two reasons. A non-rejection is not evidence that the class contains the truth (the test may simply lack power; this is the same inversion of the burden of proof that motivates the equivalence formulation), and the region in a data-selected class is not a post-selection confidence region. The first reason can be quantified.

Proposition 3 (What a lack-of-fit test cannot see)**.** *Let the truth be $F = F_{\beta^\circ} + h$, where F_{β° is the least-squares projection of F on the fitted class over the design and h lies in a larger class $\mathcal{H}$ of $m_{\mathcal{H}}$ functions per component. The F-test of the fitted class (m functions per component) against $\mathcal{H}$ has noncentrality $\lambda = N\|h\|^2_{\mu_N}/\sigma^2$, where $\|\cdot\|_{\mu_N}$ is the root-mean-square over the design, and power $\gamma(\lambda)$ given by the noncentral F distribution with $n(m_{\mathcal{H}} - m)$ and $n(N - m_{\mathcal{H}})$ degrees of freedom. Let λ_γ solve $\gamma(\lambda_\gamma) = \gamma$ and*

$$G(\xi) \;=\; \max_{h \in \mathcal{H}_\perp} \frac{\|[\xi, h]\|_{\nu,W}}{\|h\|_{\mu_N}}, \tag{7}$$

with $\mathcal{H}_\perp$ the fields of $\mathcal{H}$ orthogonal to the fitted class over the design. Then every omitted part with $\|h\|_{\mu_N} \le \sigma(\lambda_\gamma/N)^{1/2}$ is detected with probability at most γ. Each such h adds to the bracket $[\xi, F_{\beta^\circ}]$ a term with $\|[\xi, h]\|_{\nu,W} \le G(\xi)\,\sigma(\lambda_\gamma/N)^{1/2}$, so it changes the numerator of the defect of ξ by at most this amount. One such part, $h^\star$, attains the bound on $\|[\xi, h]\|_{\nu,W}$; if ξ is an exact symmetry of F_{β°, the numerator then changes by exactly this amount, and the truth $F = F_{\beta^\circ} + h^\star$ has

$$\rho_\nu(\xi; F) \;=\; \frac{G(\xi)\,\sigma\,(\lambda_\gamma/N)^{1/2}}{\big(\|DF\,\xi\|^2_{\nu,W} + \|D\xi\,F\|^2_{\nu,W}\big)^{1/2}} \tag{8}$$

while the test of the fitted class rejects with probability at most γ.

To first order in $h^\star$ the denominator of (8) is that of F_{β°. The invisible defect therefore scales like $\sigma/\sqrt{N}$, as the width of the certificate does, so whether a lack-of-fit gate protects the certificate is decided by constants: by the derivative amplification in $G(\xi)$, which grows with the degree of the omitted terms, and by the degrees of freedom of the test. Section 2.8 shows both regimes: one where the test is the more sensitive instrument, and one where misspecification that it misses with probability 0.5 can produce true defects of about 1.7 times the tolerance and the gate certifies falsely in nearly half of the replications. A valid procedure for case (ii) needs a class that is assumed to contain the truth. Given that assumption, choosing among its nested subclasses costs only a split of α (Proposition 4); the assumption itself costs power, because a larger class widens the region (Section 2.8).

Proposition 4 (Certifying simultaneously over a nested family)**.** *Under model* (3) *with a fixed design, let* $\mathcal{F}_1 \subset \cdots \subset \mathcal{F}_K$ *be nested linear model classes and assume* $F \in \mathcal{F}_K$*. Give each class the region* $\mathcal{B}_k$ *of Proposition 2(iii) at level* α/K*, computed in that class. For* $k < K$ *write* $h_k(\beta)$ *for the part of* $F_\beta \in \mathcal{F}_K$ *omitted by the least-squares projection on* $\mathcal{F}_k$ *over the design, let* $\bar{\eta}_k = \sup_{\beta \in \mathcal{B}_K} \|h_k(\beta)\|_{\nu,W}$*, and let* U_k *be the certificate computed in* $\mathcal{F}_k$ *with the* L^2 *model-error term* $\bar{\eta}_k$ *(and* U_K *the certificate in* $\mathcal{F}_K$*). Then with probability at least* $1-\alpha$*,* $\rho_\nu(\xi_\theta; F) \le \min_k U_k(\theta)$ *for every* θ*.*

The reported bound is the smallest over the family. Each $\bar{\eta}_k$ is one trust-region maximum, because $h_k(\beta)$ is linear in β. Sample splitting, choosing the class on one half and certifying on the other, removes the reuse of the data but does not make a chosen class contain the truth. Certifying in the largest class alone, at level α, is valid under the same assumption as Proposition 4. The proposition's bound is never larger than that class's certificate at level α/K; when the smaller classes do not help, it is slightly wider than certifying in the largest class at level α, because it still pays for the split of α.

(iii) Misspecification bounded from outside. When no available class contains the truth, replace F by $F_\beta + h$ and pay for h explicitly. In *supremum form*, with $W = I$, if $\sup_{\operatorname{supp}\nu} \|h\| \le \eta_0$ and $\sup_{\operatorname{supp}\nu} \|Dh\|_F \le \eta_1$, then $[\xi, h]$ moves the numerator by at most $\eta_1\|\xi\|_\nu + \eta_0\|D\xi\|_\nu$, where $\|\xi\|_\nu = \|\xi\|_{\nu,I}$ and $\|D\xi\|_\nu = (\int \|D\xi\|_F^2 \, d\nu)^{1/2}$, and each denominator block by at most one of those terms; valid for any smooth h and conservative, because it spends supremum norms. In L^2 *form*, if the truth lies in a larger class and the omitted part has $\|h\|_{\nu,W} \le \eta_2$, the numerator moves by at most $\|\Lambda_\theta\|_{\mathrm{op}}\eta_2$, where $\|\Lambda_\theta\|_{\mathrm{op}} = \sup_h \|[\xi_\theta, h]\|_{\nu,W}/\|h\|_{\nu,W}$ over h in the larger class is the largest gain of the linear map $h \mapsto [\xi_\theta, h]$, computed exactly as a matrix norm; the two denominator blocks are treated likewise. Both forms are expensive because the Lie derivative *differentiates*: a model error that is small in size need not be small in slope, and the conversion constant grows with the degree of the class the error is allowed to occupy. None of the bounds η_0, η_1 and η_2 can be verified from the data alone, so we treat them as sensitivity parameters and report the certified tolerance as a function of them, in the manner standard for unverifiable assumptions [28].

The errors.

Exact coverage of the least-squares region needs Gaussian homoskedastic errors on the right-hand side, independent of states that are themselves observed without error. Any other construction of $\mathcal{B}$ can be supplied instead, and the certificate is then exactly as valid as that construction: White's sandwich [29] for independent rows with arbitrary heteroskedasticity, asymptotically in the number of rows; the cluster-robust sandwich for independent trajectories with arbitrary dependence within each, asymptotically in the number of trajectories, which must exceed the number of coefficients for the region to be bounded in every direction, and in our experiments exceed it several times over before the coverage approaches its nominal level; the Newey–West sandwich [30] with an automatic bandwidth [31] for serially correlated errors along long trajectories. Noisy *states* are a different matter: the

regressors then carry error, least squares estimates a pseudo-true field, and no covariance estimate repairs the bias. That case is not covered (Section 2.12).

The target domain, the metric and the candidate class.

The claim is about the user's ν and W, and nothing is asserted outside ν; coverage of ν by the data enters through the design and is quantified by Proposition 6. We find connected subgroups whose generators lie in $\mathcal{V}$, excluding translations, for which the relative defect degenerates (Section 2.1). Discrete symmetries are outside this framework, as they are outside the Lie-algebraic framework generally.

Learned features.

The construction applies verbatim to any model that is linear in its parameters given its features, including the last layer of a network. The classical region is valid only if the features are fixed independently of the data used for the region, either chosen in advance (as our polynomial and dictionary features are) or learned on separate data, with sample splitting or cross-fitting [32]. Features learned on the same data need a region that accounts for that learning.

2.4 How small a violation can be certified, and from which data

Two questions bound what any procedure can do. How small a tolerance is achievable, and when can the observed region of state space stand in for the target domain?

A resolution limit.

Exact symmetry cannot be certified from noisy data: a system with zero defect and one with a defect just above zero generate almost the same data. The useful version of this statement is quantitative.

Proposition 5 (Le Cam bound on the certifiable tolerance)**.** *Consider model (3) with a fixed design whose empirical measure is μ_N, and a fixed generator $\xi = \xi_\theta \in \mathcal{V}$; write L_ξ, $M_\xi^{(1)}$, $M_\xi^{(2)}$ and M_ξ for the matrices L_θ, $M_\theta^{(1)}$, $M_\theta^{(2)}$ and M_θ of Lemma 2. Let $\varphi \in \{0,1\}$ be any rule for certifying ξ at tolerance δ whose false-certification rate is at most α, i.e. $\mathbb{P}_\beta(\varphi = 1) \le \alpha$ whenever $\rho_\nu(\xi; F_\beta) > \delta$. Let $s_{\max}(\xi) = \max_{\Delta \neq 0} \|L_\xi \Delta\| / \|F_\Delta\|_{\mu_N}$, attained at $\Delta^\star$ normalised so that $\|F_{\Delta^\star}\|_{\mu_N} = 1$. Then for any β^0 with $\rho_\nu(\xi; F_{\beta^0}) = 0$ and any $\delta < s_{\max}(\xi)/\|M_\xi \Delta^\star\|$,*

$$\mathbb{P}_{\beta^0}(\varphi = 1) \;\le\; \alpha \;+\; \frac{\sqrt{N}}{2\sigma}\, \frac{\delta\, \|M_\xi \beta^0\|}{s_{\max}(\xi) - \delta\, \|M_\xi \Delta^\star\|}. \tag{9}$$

Equivalently, attaining power γ at β^0 requires $\delta \ge 2\sigma(\gamma - \alpha)\, s_{\max}(\xi) \,/\, \big(\sqrt{N}\, \|M_\xi \beta^0\| + 2\sigma(\gamma - \alpha)\, \|M_\xi \Delta^\star\|\big)$.

The bound is non-asymptotic and its constants are computable for the design at hand, so it serves as a numerical benchmark for a single, pre-specified generator in a fixed model class. It is a lower bound obtained through Pinsker's inequality and one

two-point alternative, and we do not claim it is attained. Section 2.9 compares it with the certificate: the two share the $\sigma/\sqrt{N}$ scaling over two decades of N at a fixed model class, with a gap of a factor of about 8 that increases with the size of the model class. That comparison supports the scaling in N and σ, not optimality: neither the constant nor the dependence on the number of coefficients Q is addressed by the bound.

The price of incomplete coverage.

Suppose the data occupy a region with distribution μ while the claim is about ν. Both terms $DF_\beta\,\xi_\theta$ and $D\xi_\theta\,F_\beta$ of every bracket, and hence the bracket itself, lie in a fixed finite-dimensional space $\mathcal{R}$ of fields (for polynomial classes, the polynomial fields up to the degree of a bracket), so the worst-case discrepancy between the two measures is a pair of generalised eigenvalues.

Proposition 6 (Extrapolation factor)**.** *Let G_μ and G_ν be the Gram matrices of a basis of $\mathcal{R}$ under μ and ν, with G_ν positive definite, and set $\kappa = \lambda_{\max}(G_\mu^{-1/2} G_\nu G_\mu^{-1/2})$ and $\kappa' = \lambda_{\max}(G_\nu^{-1/2} G_\mu G_\nu^{-1/2})$, with $\kappa = \infty$ when G_μ is singular. Then for every $q \in \mathcal{R}$, $\|q\|_\nu^2 \le \kappa\,\|q\|_\mu^2$ and $\|q\|_\mu^2 \le \kappa'\,\|q\|_\nu^2$, with equality attained, and consequently, for every field F in the class,*

$$\rho_\nu(\xi; F) \;\le\; \sqrt{\kappa\kappa'}\;\rho_\mu(\xi; F) \qquad \text{for every } \xi \in \mathcal{V}, \tag{10}$$

where ρ_μ is Definition 1 with ν replaced by μ. Moreover $\kappa = \infty$ if and only if some nonzero $q \in \mathcal{R}$ vanishes μ-almost everywhere, i.e. the data lie in the zero set of a nonzero field in $\mathcal{R}$, such as a possible bracket.

As a *diagnostic*, κ and κ' are computable from the observed states alone, before any symmetry analysis. In floating point G_μ is treated as singular when its smallest eigenvalue is at most 10^{-10} times its largest; no pseudoinverse or regularisation is used, and the factor $\sqrt{\kappa\kappa'}$ is then reported as infinite. As an *explanation*, apply the proposition to the true field, assumed to lie in the fitted class: generators whose defects measured on the data are at most δ have true defects $\rho_\nu \le \sqrt{\kappa\kappa'}\,\delta$. Let $r^\star$ be the dimension of the true symmetry algebra and $\rho^\star$ the square root of the $(r^\star{+}1)$-th smallest generalised eigenvalue of the true pencil $(C(\beta^\star), D(\beta^\star))$ on ν (for a system without symmetry, its smallest true defect). By the Courant–Fischer theorem every subspace of dimension $r^\star + 1$ contains a generator with true defect at least $\rho^\star$, so declaring more generators than the true algebra has is possible only if

$$\sqrt{\kappa\kappa'} \;\ge\; \rho^\star/\delta. \tag{11}$$

This is a one-sided, population-level statement. Falling below the threshold rules out declaring more generators than the true algebra has, and any declaration for a system without symmetry. It does not rule out a declaration of the right dimension tilted towards generators with true defect between δ and $\sqrt{\kappa\kappa'}\,\delta$, nor errors from estimating the field. Exceeding the threshold is a warning rather than a prediction, since the

bound is worst-case over $\mathcal{R}$. The infinite case is not exotic: a single trajectory of the Hopf normal form converges onto its limit cycle, the unit circle, and once the transient is discarded every field in $\mathcal{R}$ divisible by x^2+y^2-1 vanishes on the data to within the eigenvalue threshold, so the factor is reported as infinite. Single trajectories of the Van der Pol and Lorenz systems give large but finite factors (Supplementary Table S3).

Additive derivative noise and the plug-in defect.

It is natural to expect that noise makes small singular values smaller. Under model (3), that is, with noise on a measured right-hand side, a correctly specified class and exactly observed states, the opposite happens near a true symmetry. Since $\mathbb{E}\|L_\theta\hat{\beta}\|^2 = \|L_\theta\beta^\star\|^2 + \mathrm{tr}(L_\theta\Sigma L_\theta^\top)$, where $\Sigma = \mathrm{Cov}(\hat{\beta})$, and likewise for the denominator, estimation error inflates *both* parts of equation (4); for a direction close to a true symmetry the numerator is near zero, so its relative inflation is unbounded while the denominator's is not. The plug-in defect of a true symmetry is therefore biased upward and all directions are pulled toward a common noise-dominated value, so a fixed tolerance loses *detections* as the noise grows. For a direction whose true defect is well above zero the argument gives no sign for the bias, since the common value may lie above or below it, and a second effect pulls downward: the generator chosen to *minimise* the plug-in defect is selected for a small value (the winner's curse), which can push the minimal plug-in defect below its true value. On a system without symmetry, whether noise produces false declarations therefore depends on how far the smallest true defect lies above δ; in our tests a threshold rule at δ declared a false symmetry in at most 3.7% of replications, and only when that defect was within a factor of about two of δ. Both effects are measured in Section 2.7, and the winner's curse at small samples and high noise in Section 2.11. The argument concerns individual directions: on a symmetric system the noise also tilts the estimated subspace away from the true algebra, and a fixed tolerance can then declare generators whose true defect exceeds it, which in our tests was the more frequent failure (Section 2.6). Proposition 1 removes the need to reason about either the winner's curse or the tilt. The bias argument above holds only under model (3), with noise on the derivative and exact states: noise in the *states* acts quite differently, by attenuating nonlinear terms, and can manufacture symmetry or hide it, depending on the system (Section 2.12).

2.5 Experimental design

Ten polynomial systems on $\mathbb{R}^2$ and $\mathbb{R}^3$ serve as the main benchmarks, four with a nonzero symmetry algebra inside the affine candidate class and six with none (Methods). Their symmetry algebras are exact rational nullspaces and the true defect of any generator is available in closed form, so **ground truth involves no tolerance and no sampling**, which matters because the quantity being estimated is itself a near-degeneracy. Every method is scored identically, by the rate of *false certification* (the declared subspace contains a generator whose true defect exceeds δ), of *detection* (the declared dimension is at least the true one) and of *recovery* (detection, with the declared subspace containing the true algebra to within 10°). The worst-setting rates of Table 2, the zero rate of SymCert and selected rates in

	false certification			symmetric systems	
method	mean	worst setting [95% CI]	settings $> \alpha$	detection	recovery
fixed threshold $\tau = 0.005$	0.005	0.633 [0.576, 0.688]	3	0.418	0.418
fixed threshold $\tau = 0.01$	0.005	0.633 [0.576, 0.688]	3	0.523	0.523
fixed threshold $\tau = 0.02$	0.005	0.633 [0.576, 0.688]	3	0.635	0.635
fixed threshold $\tau = 0.05$	0.005	0.633 [0.576, 0.688]	3	0.784	0.784
fixed threshold $\tau = 0.1$	0.017	0.633 [0.576, 0.688]	18	0.886	0.886
fixed threshold $\tau = 0.2$	0.139	1.000 [0.988, 1.000]	52	0.960	0.959
fixed threshold $\tau = 0.4$	0.551	1.000 [0.988, 1.000]	142	0.990	0.990
largest eigengap	0.650	1.000 [0.988, 1.000]	166	0.842	0.841
nullspace rank test	0.070	1.000 [0.988, 1.000]	30	0.988	0.987
split significance test	0.070	0.920 [0.883, 0.948]	37	0.844	0.221
Weyl perturbation bound	0.000	0.000 [0.000, 0.012]	0	0.000	0.000
SymCert	0.000	0.000 [0.000, 0.012]	0	0.485	0.485
SymCert + sample splitting	0.000	0.000 [0.000, 0.012]	0	0.371	0.371

Table 2 Calibration and power over 240 settings. Ten systems, sample sizes $N = 100, 200, 400, 800, 1600$ and 3200, and noise levels $0.01, 0.03, 0.1$ and 0.3; 300 replications per setting. "Mean" is the average false-certification rate over settings and "worst setting" its maximum, with an exact Clopper–Pearson interval for the Monte Carlo error of that setting; the next column counts settings whose rate exceeds $\alpha = 0.05$ (out of 240). Detection (declared dimension at least the true one; for the split-sample test, which declares at most one direction, any declaration) and recovery (detection, with the true algebra inside the declared subspace to within $10°$) are averaged over the 96 settings of the four symmetric systems. Among the thresholds examined none keeps the worst-setting rate near α; the rank and split-sample tests are valid tests of exact symmetry and still poor discovery rules; the Weyl certificate is valid but never certifies.

the text are given with exact binomial intervals for their Monte Carlo error; Methods bounds that error for every other rate. Unless stated otherwise $\delta = \alpha = 0.05$ and the region is the exact least-squares ellipsoid.

The framework we build on does not prescribe a threshold for declaring a symmetry (it establishes that the symmetries are a nullspace), so there is no single baseline, and we sweep the threshold instead of picking one, judging false certification at $\delta = 0.05$ throughout. We give the threshold rule the advantage of the *exact* target-domain Gram matrix, which isolates the statistical question from the coverage effect studied separately. The methods are (i) a fixed threshold on the plug-in defect, swept over $\tau \in [0.005, 0.4]$; (ii) the largest relative eigengap; (iii) a Wald rank test for the nullspace dimension; (iv) a split-sample significance test that selects a direction on one half and tests H_0: exact symmetry on the other, declaring a symmetry when it fails to reject; (v) a Weyl-inequality certificate, the natural singular-value perturbation answer; (vi) SymCert; (vii) SymCert with sample splitting, which chooses the plug-in directions on one half of the data and certifies them with the simultaneous certificate on the other (Methods).

2.6 Calibration and power

Table 2 and Fig. 2a summarise 240 settings (10 systems × 6 sample sizes × 4 noise levels, 300 replications each; 72,000 replications in total).

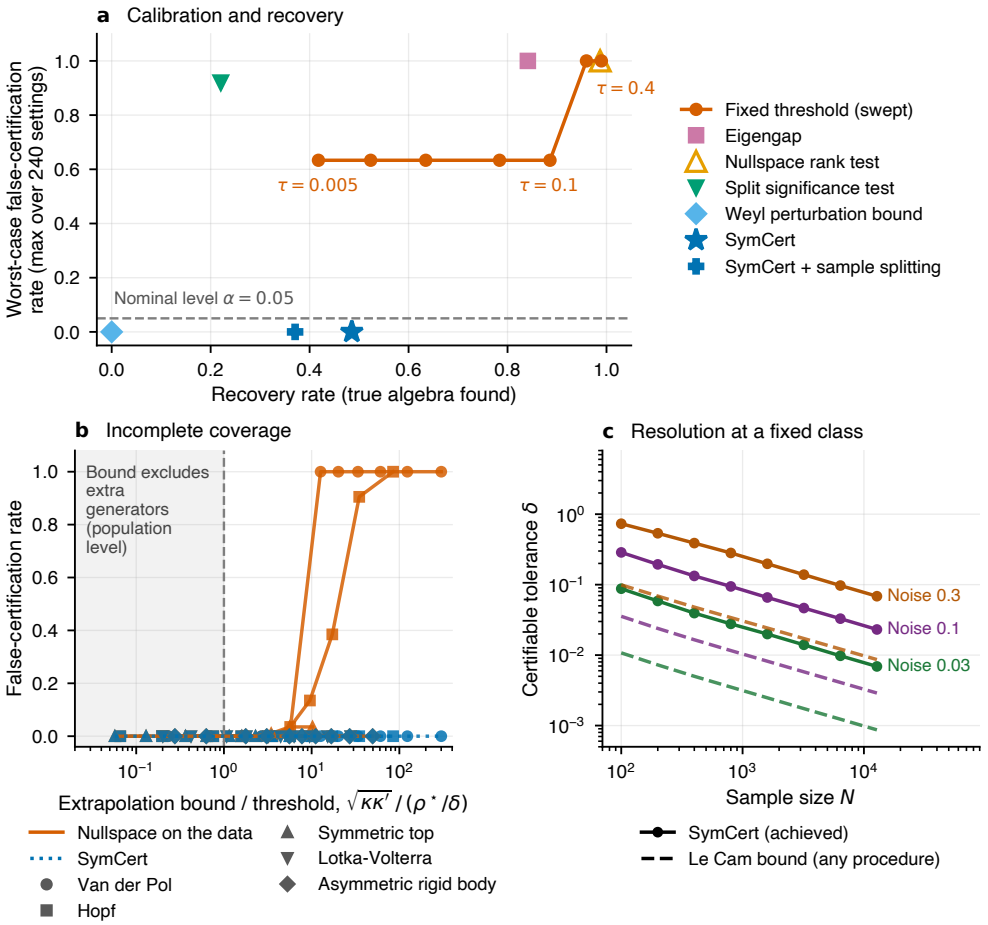


Fig. 2 Calibration, coverage and resolution. a, Every method of Table 2 placed by the rate at which it recovers the true symmetry algebra (horizontal; averaged over the four symmetric systems) against its largest false-certification rate over the 240 settings (vertical; 300 replications each). The line traces the fixed-threshold rule as its threshold τ is swept from 0.005 to 0.4 ($\tau = 0.005$, 0.1 and 0.4 labelled), with false certification always judged at the tolerance $\delta = 0.05$; none of these thresholds brings the worst case near the nominal level (dashed). The rank test (hollow triangle, on the $\tau = 0.4$ point) and the split-sample significance test are valid level-0.05 tests of exact symmetry and still poor discovery rules. SymCert and its sample-splitting variant are both at zero, the variant with lower recovery; the Weyl perturbation bound is also at zero but never certifies. **b**, Shrinking the region the data occupy while the claim stays about the full domain (50 settings, five systems, 200 replications each). The horizontal axis is the extrapolation bound of Proposition 6 (mean over replications) divided by the value $\rho^\star/\delta$ of (11); in the shaded region the bound rules out, at population level, declaring more generators than the true algebra has, and no false certification occurred in any of the 13 settings there. Orange solid: nullspace computed on the observed data; blue dotted: SymCert. Markers identify the system (circle, Van der Pol; square, Hopf; up-triangle, symmetric top; down-triangle, Lotka–Volterra; diamond, asymmetric rigid body). **c**, Median certified tolerance of the rotation generator for the exactly symmetric member ($\varepsilon = 0$) of the broken-Hopf family $F_\varepsilon = F_{\mathrm{Hopf}} + \varepsilon x^2 \partial_x$ (solid with markers; degree-3 class, 200 replications per point; colours give the noise level) against the non-asymptotic Le Cam bound of Proposition 5 at power 0.5 evaluated on each design (dashed). Both scale as $\sigma/\sqrt{N}$; the gap is a factor of about 8 at this model class and grows with the class (Supplementary Table S4).

SymCert *never certified a false symmetry.* In none of the 72,000 replications did the certified subspace contain a generator whose true defect exceeds the tolerance. With 300 replications per setting this bounds the false-certification rate of every setting by 0.010 (one-sided 95%), and simultaneously over all 240 settings by 0.028. The guarantee itself comes from Proposition 1; the simulation checks the implementation and shows how far below α the realised rate is.

None of the non-trivial thresholds we examined controls the error. A threshold of zero, or one so small that it never certifies, controls false certification trivially and has no power; among the thresholds from 0.005 to 0.4 on this grid, every one has a worst-setting rate of at least 0.63 (95% interval 0.58 to 0.69), even though the average over settings is as low as 0.005 for small thresholds. Lowering τ does not help: at $\tau = 0.005$ detection has already fallen to 0.42. For every threshold up to 0.1 the worst setting is the planar rotation at the highest noise level with $N = 100$, and for thresholds up to the tolerance the only settings above α are that system at that noise level with $N = 100$, 200 and 400 (rates 0.63, 0.34 and 0.12). There the threshold declares the right number of generators, and the declared subspace even lies within 10° of the true algebra, so the error counts as a recovery; but the noise has tilted it just enough to contain generators whose true defect exceeds the tolerance, because the defect grows linearly with the tilt. The remaining errors at $\tau = \delta$, at rates of at most 0.04, also occur on symmetric systems at noise levels 0.1 and 0.3, where a tilted subspace of too small a dimension can contain such a generator as well. Thresholds above the tolerance fail more simply: they certify, on systems without symmetry such as the Lorenz system, directions whose defect lies between δ and τ. Averaged over the symmetric settings, recovery and detection therefore differ by less than 0.002 for every rule in Table 2 except the split-sample test, and not at all for SymCert. That test declares at most one direction, which is counted as a detection whenever it is declared; it detects at a rate of 0.84 but recovers at only 0.22, because a single direction can neither contain a two- or three-dimensional algebra nor be relied on to point along a one-dimensional one.

Valid tests are not safe either. The rank test and the split-sample significance test are correct level-0.05 tests of "this is an exact symmetry". Used as discovery rules they reach worst-setting false-certification rates of 1.00 and 0.92 (mean rate 0.070 for each). This is the inverted burden of proof made quantitative: their power is lowest exactly where the data are least informative, and low power is read as evidence for symmetry.

The tolerance is a choice. Repeating the comparison across $\delta \in [0.01, 0.4]$ on a smaller grid (six of the systems, $N = 400$ and 3200, noise levels 0.03 and 0.1; Supplementary Fig. S4), SymCert's worst-case error is zero at every tolerance, and its detection rate on the three symmetric systems climbs from 0.25 at $\delta = 0.01$ to 1.00 at $\delta = 0.2$: with data of this quality, a user who will accept a 20% relative defect can have both error control and full detection, and one who insists on 1% cannot. The grid leaves out the small samples and the highest noise level, where detection at $\delta = 0.05$ is lower (Fig. 3a).

Validity is not free, and neither is it vacuous. SymCert detects less often than a threshold at the same tolerance (0.485 against 0.784; for both rules recovery equals

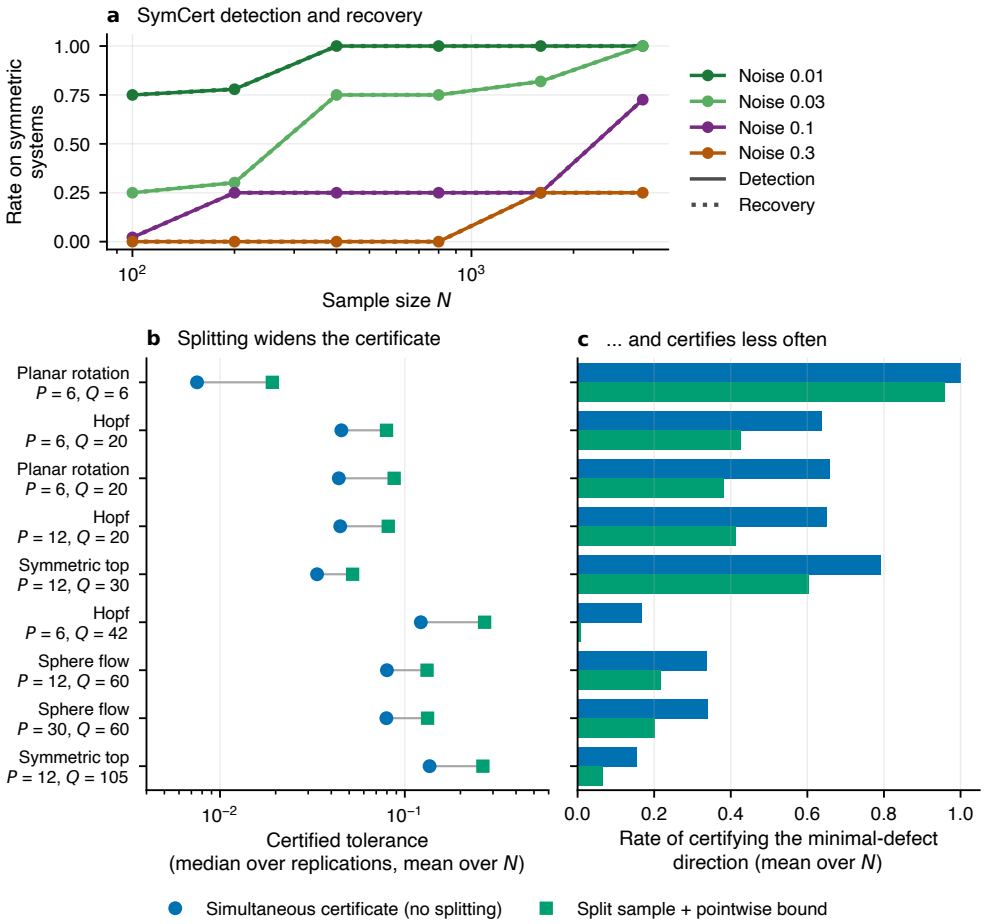


Fig. 3 The price of validity, and whether to split the sample. a, Detection (solid) and recovery (dotted, coinciding here) of the true algebra by SymCert on the four symmetric systems, against sample size, for four noise levels (colours): at noise level 0.01 the full algebra is found in every replication from $N = 400$; at noise level 0.3 it is not, because the data do not support the claim. **b**, Certified tolerance of the simultaneous certificate (blue circles) and of the split-sample pointwise bound (green squares) in nine model configurations, one row each, ordered by the number of model coefficients Q (noise level 0.05; median over 300 replications at each of $N = 200, 400, \ldots, 6400$, averaged over the six sample sizes). **c**, Rate at which each rule certifies the minimal-defect plug-in direction, averaged over the same six sample sizes. Both rules certify that single direction only, so in the six configurations whose true algebra has dimension 2 or 3 this is not detection in the sense of Methods. The simultaneous certificate is narrower, by factors of 1.6 to 2.6, and certifies that direction at least as often in all nine: splitting protects against a selection effect that simultaneity already removes, and pays for it with half the data.

detection). Figure 3a shows where the cost falls: at noise level 0.01 detection reaches 1.00 by $N = 400$; at noise level 0.3 it does not, because the data at that noise level do not support the claim. The Weyl certificate shows the alternative failure: valid, and never certifying anything.

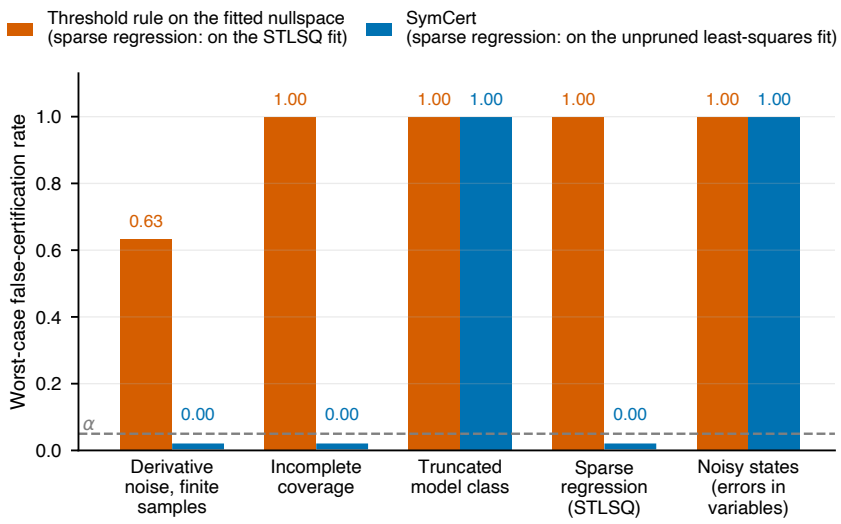


Fig. 4 Where false symmetries come from. Largest false-certification rate, at tolerance $\delta = 0.05$, of a threshold rule on the fitted nullspace (orange) and of SymCert (blue) under five mechanisms: additive noise on the derivative with finite samples (240 settings; the threshold's errors all occur on symmetric systems at noise levels 0.1 and 0.3, where the noise tilts the declared subspace); incomplete coverage of the target domain (settings with the data in at most 30% of the box width); a model class too small to contain the truth (linear or quadratic fits to cubic systems); sparse regression that deletes a weak symmetry-breaking term (threshold 0.25); and noisy states (hardening Duffing oscillator, state-noise standard deviation up to 0.12). SymCert controls the first two by construction. Its guarantee does not cover the truncated class or noisy states, because its validity is conditional on a class that contains the truth and on exactly observed states. In the sparse-regression group the orange bar is the threshold rule on the STLSQ fit and the blue bar is SymCert on the unpruned least-squares fit, which certified nothing false; certifying from the pruned model would rest on a class that no longer contains the truth. A zero rate is drawn as a short bar; the dashed line marks $\alpha = 0.05$.

2.7 Noise, coverage, and the model class

Figure 4 isolates the candidate mechanisms.

Additive derivative noise. In the calibration study, where the model class is correct, the states exact and the data cover the target domain, the threshold rule at the tolerance never certified falsely on a system without symmetry. We push this hard: four systems with no symmetry (Van der Pol, Lotka–Volterra, Duffing and the asymmetric rigid body, with smallest true defects of 0.21 to 0.29, at least four times the tolerance), noise levels 0.1, 0.3, 0.5 and 1.0, $N = 30$, 60, 120 and 400, 500 replications per cell, 32,000 in all. On Van der Pol, Duffing and the rigid body the median smallest plug-in defect rises with the noise, to between 2.0 and 2.4 times its true value at noise level 1.0 and $N = 30$ (pooled over systems and sample sizes, 1.01, 1.09, 1.27 and 1.64 times at the four noise levels); on Lotka–Volterra it stays within 1.00 to 1.14 times, with 31% to 59% of replications below the true value. The smallest value observed, 0.066 (Lotka–Volterra, true value 0.271), was still above the tolerance, which reflects the margin to δ rather than an upward bias; the argument for an upward bias holds only

for directions near a true symmetry (Section 2.4). A second study on the same four systems, at noise level 0.2 with N doubling from 100 to 6400 and 500 replications per cell, measures the competing selection effect, which at this noise level is small and decays with N: at $N = 100$ the mean of $\rho_\nu(\hat{\xi}; F_{\beta^\star}) - \rho_\nu(\hat{\xi}; F_{\hat{\beta}})$ for the selected direction is $+0.0150 \pm 0.0017$ on Van der Pol, $+0.0078 \pm 0.0004$ on Lotka–Volterra, $+0.0060 \pm 0.0021$ on Duffing and -0.0073 ± 0.0009 on the asymmetric rigid body (means $\pm$ standard errors), and the Van der Pol gap falls to 0.0002 ± 0.0002 at $N = 6400$. With fewer samples and more noise it is larger, up to 0.15 on average in the selected-generator study of Section 2.11.

A false symmetry requires the noise to carry the smallest defect down to δ, so a third study uses systems near the tolerance: broken-Hopf members whose smallest true defect is 1.2, 1.5, 2 or 4 times δ, and the hardening Duffing oscillator of the state-noise study below (0.102), on $[-1, 1]^2$, with $N = 100$, 400 and 1600, noise levels 0.05, 0.1, 0.2 and 0.3 and 300 replications per setting (18,000 in all). No generator is within the tolerance, so every declaration is false. The threshold rule at the tolerance declared one in up to 3.7% of replications at 1.2δ ($N = 100$, noise level 0.05; 95% interval 1.8% to 6.5%), at most 0.7% at 1.5δ, 0.3% at 2δ and on the hardening Duffing oscillator, and never at 4δ; SymCert certified nothing in any of the 18,000 replications. The smallest plug-in defect fell below its true value in up to 52% of replications, and at high noise and small N it was inflated (median up to 3.7 times the true value). Additive noise therefore does create false symmetries near the tolerance, but rarely in these tests, and only when the smallest true defect is within about a factor of two of δ; the dominant noise-driven failure of a threshold rule is the tilt of the estimated algebra of a symmetric system.

Coverage.

We shrink the region the data occupy from the full target box to a box of 0.15 times the half-width, keeping $N = 2000$ and noise level 0.02, and run the nullspace computation on the empirical measure of the data, which is what one does when the sample is taken to stand for the domain of interest. Figure 2b plots the false-certification rate against the extrapolation bound $\sqrt{\kappa\kappa'}$ (mean over replications) divided by the value $\rho^\star/\delta$ of (11). Of the 50 settings, 13 fall below this threshold, where Proposition 6 rules out, at population level, declaring more generators than the true algebra has (any declaration, for the six settings of systems without symmetry). In all 13 the observed rate is exactly zero; this is an observation, not a consequence, since the bound concerns the noise-free field and does not exclude a tilted declaration of the right dimension. Of the 37 beyond the threshold, 14 (six on Van der Pol, five on Hopf and three on the symmetric top) do produce false certifications, rising to a rate of 1.0. SymCert, which measures the defect on the target domain and inherits the design's uncertainty through $\hat{\Sigma}$, never certifies falsely and instead reduces the dimension of its certified subspace to zero as coverage degrades: on the Hopf system it certifies the true one-dimensional algebra when the data cover the domain and almost never once the data occupy less than half of it. On trajectory data (Supplementary Table S3) a single trajectory of the Hopf system converges to the unit circle, an algebraic curve, so some bracket vanishes on the data

to within the eigenvalue threshold and the computed extrapolation factor is infinite; single trajectories of the Van der Pol and Lorenz systems, whose attractors are not such curves, give large but finite factors (34.9 and 231.3), eight Hopf trajectories bring the factor down to 24 and a uniform design to 1.1.

The truncated model class.

Fitting a linear or quadratic model to four cubic systems (design in Methods) produces a fitted field with exact or approximate symmetries the system does not have (Fig. 4). An affine field $Ax + c$ always has a symmetry algebra of dimension at least n in the affine class (exactly n for a generic fit); the threshold rule declares these in every replication and SymCert certifies them in up to 100%. A quadratic fit is less constrained, but its smallest plug-in defect still falls within the tolerance often enough that the threshold rule declares a false symmetry in 70% to 100% of replications; SymCert certifies one in a single setting (Van der Pol, $N = 1600$). This is a failure of the model-class assumption, not of the certificate. In these gross truncations a lack-of-fit F-test against the correct class rejects in 100% of replications, so gating on it would have prevented every false certification here; the next subsection shows that this does not generalise.

The same failure arises without anyone choosing to truncate. Sequentially thresholded least squares, the sparse regression at the core of data-driven model discovery [16, 33] and the object of its own convergence theory [34], deletes small coefficients. On the broken-Hopf family $F_\varepsilon = F_{\mathrm{Hopf}} + \varepsilon x^2 \partial_x$ (design in Methods), whose rotation defect is 0.064 at $\varepsilon = 0.1$, above the tolerance, a sparsity threshold of 0.25 deletes the breaking term, after which the threshold rule (plug-in defect at most δ) declares the rotation a symmetry in 100% of replications, against 17% for the same rule on the unthresholded least-squares fit. Certifying from the unpruned least-squares fit gives zero.

2.8 Choosing the model class from the data

Here the truth is the Hopf normal form plus $\varepsilon P_k(x)\,\partial_x$, with P_k the Legendre polynomial of degree $k \in \{5, 7\}$. Because P_k is orthogonal to every polynomial of lower degree on the target box, a degree-3 fit sees (up to sampling error) the exactly symmetric Hopf field, while the true rotation defect grows linearly in ε; the family of degrees $\{3, k\}$ contains the truth in its largest member. Figure 5 compares five procedures at noise level 0.1 (Methods).

When the omitted term has degree 5 the lack-of-fit test is the more sensitive instrument. At $N = 3200$ a misspecification that it detects with probability only 0.5 can produce a true rotation defect of about 0.039 (to first order in the omitted part), below the tolerance, and the gate never certified falsely when the rotation defect differed from the tolerance (Fig. 5a); with the rotation defect exactly δ it certified, in 6% of replications at $N = 3200$ (95% interval 3% to 9%), a direction whose true defect lay just above δ. When the omitted term has degree 7 the bracket amplifies it more and the test spreads its power over more degrees of freedom: misspecification invisible at power 0.5 can produce defects of about 0.087, about 1.7 times the tolerance, and at $N = 3200$ the gate certifies falsely in 46% of replications (95%

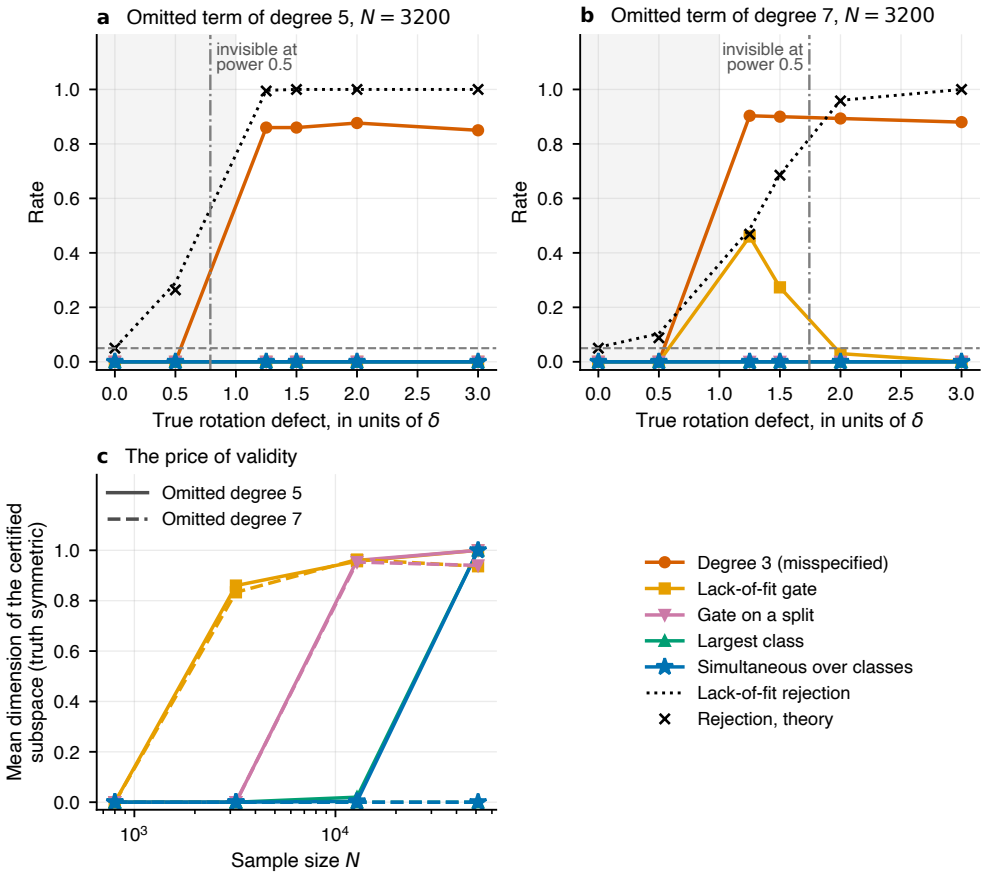


Fig. 5 Choosing the model class from the data. The truth is the Hopf normal form plus a Legendre term of degree 5 (**a**) or 7 (**b**) that a degree-3 fit cannot see; noise level 0.1, $N = 3200$, 300 replications per point. Lines with markers give the false-certification rate of five procedures (certifying in degree 3; the lack-of-fit gate; the gate applied to one half of the data and certification on the other; the largest class; certificates simultaneous over both classes), where curves at zero overlap; the dotted line gives the observed rejection rate of the lack-of-fit test, and black crosses give its power from the noncentral F distribution. The shaded region is where certifying is correct (true defect at most δ); the setting with a rotation defect exactly equal to δ is omitted from the plot. The dash-dotted line marks the true defect, to first order in the omitted part, that a misspecification detected with probability 0.5 can produce (Proposition 3). **c**, Mean dimension of the certified subspace when the truth is exactly symmetric, as a function of N, for the two valid procedures (largest class; simultaneous over both classes) and, for comparison, the lack-of-fit gate and the gate on a split (colours as in **a** and **b**; solid: omitted term of degree 5; dashed: degree 7). All 56 settings are in Supplementary Table S5.

interval 40% to 52%) at a rotation defect of 1.25δ, in 27% (22% to 33%) at 1.5δ and in 39% (33% to 45%) at exactly δ, precisely in the range that Proposition 3 marks out (Fig. 5b). The observed rejection rates agree with the noncentral-F power throughout (largest difference 0.049, less than two Monte Carlo standard errors). Choosing the class on one half of the data and certifying on the other does not remove the problem: it certifies falsely in up to 11% of replications (8% to 15%) when the rotation defect

exceeds δ (degree 7, $N = 12{,}800$, 1.25δ) and in 19% (15% to 24%) when it equals δ (degree 7, $N = 12{,}800$), because a class chosen by a test with half the data is more often too small. The two valid procedures, certifying in the largest class and certifying simultaneously over both classes, never certified falsely in any of the 56 settings. Their price is power (Fig. 5c): when the truth is exactly symmetric they certify almost nothing below $N = 51{,}200$ (mean dimension of the certified subspace at most 0.02) if the largest class has degree 5, and nothing at all in the range studied if it has degree 7, whereas the gate certifies the rotation from $N = 3200$. This power is the price of assuming the larger class, which spares one from trusting a test that failed to reject; choosing among the nested classes adds only a split of α (Proposition 4).

2.9 Certified tolerance against the Le Cam bound

Figure 2c tracks the certified tolerance of the rotation generator for the exactly symmetric member ($\varepsilon = 0$) of the broken-Hopf family, whose rotation defect grows continuously from zero with ε (the broken members are used below), against the bound of Proposition 5 evaluated on the design of each replication. Over two decades of N and three noise levels the certified tolerance follows the $\sigma/\sqrt{N}$ scaling of the bound (fitted log–log slopes between -0.49 and -0.52), exceeding it by a median factor of 8.0 at each of the noise levels 0.03, 0.1 and 0.3 in the degree-3 class ($Q = 20$). In this class the factor is nearly constant in N and σ, but it is not constant across model classes: fitting the same symmetric system in classes of degree 5 and 7 ($Q = 42$ and 72) raises it to 10.9 and 13.2 (medians over $N = 800$ to 12,800, where every bound is finite and below 1; Supplementary Table S4). The pointwise bound, valid here because the rotation is fixed in advance, gives median factors of 8.0, 10.7 and 12.9 over the same range; it is much wider at small N (infinite for degree 7 at $N \leq 200$) and slightly narrower from $N = 3200$ on (at $N = 12{,}800$, factors of 7.7, 10.1 and 11.6 against 7.9, 11.0 and 13.2). The growth of the gap with Q is therefore shared by a bound that is not simultaneous, so it is not mainly a cost of simultaneity. It reflects the model class itself, since the bound for one generator depends on every coefficient that enters its bracket, together with the looseness of the two-point lower bound, and we cannot apportion it between the two. The comparison therefore supports the scaling in N and σ at a fixed model class; it does not show that the constant, or the dependence on Q, is optimal.

The same experiment measures the opposite claim. At $N = 1600$ and noise level 0.1, the lower certificate rules out exact symmetry ($L > 0$) for 49.5% of replications when the true rotation defect is 0.013 and for 100% when it is 0.032; when the system really is symmetric it does so in 1.0% of replications (2 of 200; 95% interval 0.1% to 3.6%), below the nominal 5%.

2.10 Problem dimension, and whether to split the sample

The width of the simultaneous certificate is governed by the number of model coefficients Q, through the radius of the confidence ellipsoid; the width of the split-sample pointwise bound is governed by an effective dimension of the generator being tested, but it sees half the data. Figure 3b,c settles which wins here: **the**

simultaneous certificate is narrower in all nine configurations, by factors of 1.6 to 2.6, and it certifies the minimal-defect plug-in direction at least as often in all nine (both averaged over $N = 200$ to 6400). Sample splitting, the standard remedy for the selection effect, is counterproductive here, because the selection effect it protects against is already handled by simultaneity. Increasing the *candidate generator* class is nearly free: moving from the 6-dimensional affine class to the 12-dimensional class of all quadratic generators on the Hopf system changes the certified tolerance (median over replications, averaged over $N = 200$ to 6400, as in Fig. 3b) from 0.0454 to 0.0447. Increasing the *model* class is not: raising the polynomial degree from 3 to 5 (Q from 20 to 42) widens it from 0.045 to 0.122. Search widely over generators, and spend the sample-size budget on the model class.

2.11 Is the guarantee tight?

A certificate that certified nothing would also never certify anything false, so we measured how conservative ours is (Supplementary Fig. S5). The coefficient region covers the true coefficients at close to its nominal level, as Proposition 2 says it should: pooled over the six settings, within 0.02 of nominal at every level from 0.50 to 0.99 (0.94 to 0.96 in every setting at nominal 0.95); single settings, with 80 replications each, scatter more widely, from 0.35 to 0.59 at nominal 0.50, within Monte Carlo error over the 36 setting-level combinations. The certificate interval contains the true defect of every plug-in direction in all replications at every nominal level from 0.50 to 0.99. Each bound is a worst case over a region for all Q coefficients, and the pointwise bound for a generator fixed in advance is nearly as wide (Section 2.9), so this conservatism is not mainly a cost of simultaneity over generators.

To see what a finite-sample guarantee costs, we compared the certificate with a calibration-*targeting* alternative, the parametric bootstrap that draws $\beta^* \sim N(\hat{\beta}, \hat{\Sigma})$ and reports the α and $1-\alpha$ quantiles of $\rho_\nu(\xi_\theta; F_{\beta^*})$ [35], on the rotation generator of the broken-Hopf family. Its upper bound is 5% to 42% lower than the certificate's at $\alpha = 0.05$ and up to 47% lower at $\alpha = 0.2$, the gap being largest at a true symmetry and shrinking with N when the defect is not zero. For this generator, fixed in advance, the bootstrap upper bound held: with the true defect just above the tolerance (1.02δ) and sample sizes from 800 to 51,200, it certified falsely in at most 15% of replications at nominal $\alpha = 0.5$ (95% interval 11% to 20%), 4.3% at 0.2 (2.3% to 7.3%) and 0.7% at 0.05 (0.1% to 2.4%), all within its nominal levels, while SymCert never did. Near a symmetry, where the defect, a norm, is not a smooth function of the coefficients and its plug-in value is biased upward, the bootstrap fails in its lower quantile, which lies above the true defect: its interval covered the truth in at most one of 200 replications at either level ($\alpha = 0.05$ and 0.2, nominal 90% and 60%) whenever the true defect was 0.01 or less, and at 0.02 ($N = 3200$) in 36% (29% to 43%) against a nominal 90% and 8.5% (5.0% to 13.3%) against 60%, so it would wrongly exclude exact symmetry. The certificate's coverage never fell below 0.93 (nominal 0.80) or 0.985 (nominal 0.95). The bootstrap upper bound itself fails for a generator selected from the data (Supplementary Fig. S5e). On four systems without symmetry (Van der Pol, Duffing, Lotka–Volterra and Sel'kov, all on the box $[-1, 1]^2$, smallest true defects 0.25 to 0.64; $N = 50$ to 400, noise levels 0.2 and 0.5, 300 replications per

setting), with the generator of smallest plug-in defect bounded as if it had been fixed in advance, its coverage was below nominal in 9 of 32 settings at nominal 0.80 and in 8 of 32 at 0.95, with the whole 95% interval below nominal in 5 settings at each level, nine of these ten on the Duffing oscillator (worst 0.59, 95% interval 0.54 to 0.65, and 0.90, 0.86 to 0.93, at $N = 50$ and noise level 0.5): selection biases the plug-in defect downward, by up to 0.15 on average, and a bootstrap centred at it does not see the selection. The certificate's upper bound contained the true defect in every replication. With true defects far above δ, the failure shows as an understated tolerance, not as a false certification at 0.05. The extra width is therefore the price of a guarantee that holds by theorem, in finite samples, simultaneously over generators and for both bounds, and the experiment with a data-selected generator, on the Duffing oscillator in particular, shows a case in which a bound without this guarantee fails.

2.12 Departures from the assumptions

Table 3 separates the two coverages that matter: that of the coefficient region, the event on which every certificate rests, and that of the certificate itself. With independent rows, the exact least-squares region keeps close to its nominal level under Student-t_4 and Laplace errors (at least 0.938) but not under heteroskedastic errors (0.82), and White's sandwich undercovers at the smaller sample size (0.77 to 0.88 at $N = 400$, a known small-sample weakness of the plain sandwich with a chi-square reference) and approaches its nominal level at $N = 3200$ (0.93 to 0.96). The certificate covered the true defect of every plug-in direction in every one of these replications and never certified falsely. Its own conservatism absorbed the region's shortfall here, but that margin is not part of any guarantee, and the region is what should be chosen to match the error structure.

On trajectory data the rows are dependent through the design, but with exact states and independent errors the regression is still a correctly specified least-squares problem, and White's region covers (0.90 to 0.98). The cluster-robust region, which also allows dependent errors within a trajectory, covers less well with 48 trajectories (0.76 to 0.91): its covariance estimate has about as many degrees of freedom as there are trajectories, here only 1.6 to 2.4 times the number of coefficients ($Q = 30$ or 20), even in the jackknife form that corrects the leverage bias of the plain cluster sandwich. With noisy states and central differences inside each trajectory, state noise of 0.05% to 0.5% of the box half-width barely biases the fit; differencing multiplies it into large derivative noise, which costs detection, and makes neighbouring errors negatively correlated, which the cluster-robust region accounts for and White's overestimates.

Genuinely dependent errors are another matter (Supplementary Fig. S1 and Supplementary Table S9). With errors following an AR(1) process along each trajectory, the OLS and White regions, which assume independent errors, contained the true coefficients in at most 7% of replications once the autocorrelation ϕ reached 0.5, and for $\phi \geq 0.9$ the certificate built on them failed as well, its interval covering the true defects in only 1% to 27% of replications; in these two systems the failure happened not to produce a false certification, but nothing guarantees that. The cluster-robust region, designed for this case, covered in 73% to 92% of replications with 48 trajectories and 91% to 95% with 192, and the certificate built on it covered

setting	region	region coverage	certificate coverage	false cert.
Independent rows, error distribution varied				
Gaussian	OLS (exact)	0.943	1.000	0.000
	White	0.770	1.000	0.000
Student t_4	OLS (exact)	0.940	1.000	0.000
	White	0.823	1.000	0.000
Laplace	OLS (exact)	0.938	1.000	0.000
	White	0.792	1.000	0.000
heteroskedastic Gaussian	OLS (exact)	0.823	1.000	0.000
	White	0.797	1.000	0.000
48 trajectories				
exact states	White	0.900	1.000	0.000
	cluster-robust	0.760	1.000	0.000
noisy states, differenced	White	1.000	1.000	0.000
	cluster-robust	0.760	1.000	0.000
AR(1) errors, 192 trajectories				
$\phi = 0.9$	OLS (exact)	0.000	0.155	0.000
	White	0.000	0.150	0.000
	cluster-robust	0.910	1.000	0.000
	Newey–West	0.400	1.000	0.000
$\phi = 0.98$	OLS (exact)	0.000	0.010	0.000
	White	0.000	0.010	0.000
	cluster-robust	0.910	1.000	0.000
	Newey–West	0.135	1.000	0.000

Table 3 What the certificate inherits from the coefficient region. Nominal coverage 0.95. Region coverage is the fraction of replications in which the coefficient region contains the true coefficients; certificate coverage the fraction in which the interval contains the true defect of every plug-in direction; both are minima, and the false-certification rate a maximum, over the settings in each row (independent rows: three systems, each at (N, noise level) = $(400, 0.1)$ and $(3200, 0.03)$, 400 replications each; trajectories: three systems, with exact states or with state noise of standard deviation 0.0005, 0.002 and 0.005 (0.05% to 0.5% of the box half-width), 100 replications each; AR(1): two systems, 200 replications each). Each trajectory starts from a uniform initial condition in the box and runs for 100 steps of 0.01; only its part inside the box is kept (at most 100 points), and clusters and Newey–West groups are trajectories. With noisy states the regressors carry error and no region is guaranteed to cover; the state noise here is small, so the bias is negligible and the differenced derivatives dominate the error (main text). With serially correlated errors only the cluster-robust and Newey–West regions are designed to cover.

in every replication; with 12 trajectories, fewer than the coefficients, it leaves directions unidentified and certifies nothing. The Newey–West region, whose justification is asymptotic in the length of each series, was not adequate for trajectories of at most 100 points: for $\phi \geq 0.5$ its coverage was 0.14 to 0.80 with 192 trajectories, 0.01 to 0.81 with 48 and 0 to 0.46 with 12, and it fell short already at $\phi = 0.5$, where the correlation time is only a few steps (0.75 to 0.80 with 192). The certificate built on it covered in at least 98% of replications in every other setting, but with 12 trajectories at $\phi = 0.98$ in only 80% (Hopf) and 91% (rigid body), though without a false certification. Dependent data therefore need both the right estimator and enough independent trajectories.

Noise in the states is not covered, and it can manufacture symmetry (Supplementary Fig. S3). We observed the states with independent noise of standard deviation σ_x and the right-hand side separately with small independent noise. Least squares then estimates a pseudo-true field whose nonlinear terms are attenuated. For the hardening Duffing oscillator $\dot{x} = y$, $\dot{y} = -x - 0.3x^3 - 0.2y$, whose smallest true defect is 0.102, state noise of $\sigma_x = 0.08$ lowers the smallest defect of the pseudo-true field to 0.038. As data accumulate the region concentrates on that field, not on the truth (it contained the true coefficients in none of the replications), and SymCert certified a false symmetry in every replication at $N = 25{,}600$ and 204,800; at $N = 3200$ it was still too wide to certify. The threshold rule did so at every sample size. State noise of 0.04 or 0.12 left the pseudo-true defect above the tolerance and produced no false certification, and on the symmetric Hopf system state noise of 0.08 or more destroyed detection instead. Whether errors in the states inflate or deflate the apparent symmetry depends on the system, which is why the guarantee cannot extend to them without a model of those errors.

Supplementary Fig. S2 turns to systems that no polynomial class contains (one data set of $N = 40{,}000$ points per system and degree, noise level 0.02). On a non-polynomial planar flow with an exact rotational symmetry, on the box $[-1, 1]^2$, a degree-5 fit gives an upper bound of 0.050 if model error is ignored, a bound with no guarantee. Paying for the error in L^2 against a degree-9 class, with its size taken from the known truth (an oracle, unavailable in practice), gives 0.65, and paying in supremum norm 2.65. Bounding the omitted part from the data instead gives 0.68, close to the oracle L^2 value. Both L^2 values assume that the omitted part lies in the degree-9 class, which for this flow holds only approximately, so they are sensitivity values rather than guaranteed bounds; only the supremum-norm value is a guaranteed bound, provided its oracle sizes bound the true suprema (here they are maxima over 4×10^5 points, which can fall slightly short of them). All three are vacuous at $\delta = 0.05$. Raising the fitted degree to 9 shrinks the relative L^2 model error to 0.5% and the data-driven bound to 0.27, still too wide to be useful: the bracket differentiates, so the constant converting a bound on $\|h\|$ into a bound on $\|[\xi, h]\|$ grows with the degree of the class faster than the error falls. On the damped pendulum, which has no continuous symmetry, the same accounting is cheap (true defect 0.204, bounds between 0.23 and 0.39 across degrees and forms, with the same caveat for the L^2 forms) because a degree-5 polynomial reproduces sin on the target domain to a relative L^2 error of 0.07%.

When a dictionary that contains the truth is available, no model-error term is needed (Supplementary Table S6). We fitted the same planar flow, with the unit disc as target domain, with quadratic polynomials and four radial functions, with and without the symmetry-breaking term $\varepsilon x^2 \partial_x$, and the damped pendulum with a polynomial and trigonometric library. Across 45 settings (300 replications each) SymCert never certified falsely, while a threshold at the tolerance did so in up to 6.3% of replications. With the exact rotational symmetry present, it was certified in every replication at noise level 0.03, with a median upper bound of 0.008 at $N = 3200$; the polynomial fits above, on the box $[-1, 1]^2$ with $N = 40{,}000$ and noise level 0.02, gave only vacuous bounds once model error was paid for. With a breaking term

giving the rotation a defect of 0.024 SymCert certified that approximate symmetry from $N = 800$ at the same noise level, and with a defect of 0.096 it certified nothing. On the pendulum the lower certificate excluded every 0.05-approximate symmetry in all replications at $N = 3200$ for noise levels up to 0.1, and the upper bound on the minimal defect (0.213 at noise level 0.03) was within 5% of its true value (0.204).

2.13 Scaling, units and closure

The state dimension enters twice: through the number of model coefficients Q, which sets the radius of the confidence region, and through the number of candidate generators P, which sets how many directions are searched. On linear systems in $n = 2, \ldots, 6$ built from rotation blocks, whose symmetry algebra is the commutant of the coefficient matrix and has dimension n, false certification is zero in all 20 cells (Supplementary Table S7; five dimensions, $N = 200$, 800, 3200 and 12,800) while the search space $P = n^2$ grows from 4 to 36. What the growth costs is width: at $N = 12{,}800$ the median upper bound of the minimal-defect plug-in direction rises from 0.0017 at $n = 2$ ($Q = 6$) to 0.0050 at $n = 6$ ($Q = 42$), a factor 2.9, somewhat more than the $\sqrt{Q}$ factor of 2.6. Because P and Q grow together in this design, it does not show which of the two the width depends on; at a fixed model class, enlarging the candidate class barely changes the width (Section 2.10). The dimension of the certified subspace reaches n (detection) in every replication from $N = 800$ at $n = 4$ and from $N = 3200$ at $n = 6$, and one certified plug-in subspace costs between 0.03 and 0.11 CPU-seconds across this range.

Units matter exactly as Lemma 1 predicts (Supplementary Table S1). Re-expressing one coordinate of the broken-Hopf system in units c times larger, for c from 0.01 to 100, moves the Euclidean defect of the rotation between 0.051 and 0.076 (it is 0.064 in the original units), and lowers the smallest Euclidean defect in the class for Lotka–Volterra from 0.27 to 0.0035; with the nondimensional metric neither number moves. Certifying in the rescaled Lotka–Volterra coordinates with the Euclidean metric at $N = 12{,}800$ certified a generator in every replication, and every one of those certifications was false when judged by the unit-free defect of the system; with the declared metric nothing was certified.

Certified subspaces of dimension two or more were close to closed under the bracket (Supplementary Table S2). For the radial sphere flow, whose symmetry algebra is the non-abelian $\mathfrak{so}(3)$, the certified subspace had dimension three in every replication except at the smallest sample size with the larger noise, the part of each commutator outside it was at most 0.4% of the size of its two terms, and every commutator was itself certified. For the two abelian algebras the commutators were negligible.

2.14 A physically expected symmetry in ocean measurements

Near-inertial currents in the upper ocean follow, to first order, the damped slab model $\dot{u} = fv - ru$, $\dot{v} = -fu - rv$ plus wind forcing [36], where (u, v) is the horizontal velocity and f the local Coriolis parameter. With time measured in units of $1/f$ the right-hand side is $(v - (r/f)\,u,\ -u - (r/f)\,v)$, a uniform damping plus a

quarter-turn rotation of the velocity, both of which commute with the rotation generator: the dynamics are rotation invariant. We asked whether hourly velocities of drogued drifters of the Global Drifter Program [37, 38] certify that symmetry, using 46,733 velocity–acceleration pairs from 31 drifters in the North Pacific (30–45°N, 180–140°W, June to September 2016–2019) after isolating the near-inertial band (Methods, Supplementary Table S8). The fitted linear field is $\dot{u} = 0.994\,v$, $\dot{v} = -1.028\,u$ in units of f, the textbook inertial oscillation, and the plug-in defect of the rotation generator is 0.025. Because rows along a trajectory are serially dependent, we use the cluster-robust region over drifters, which needs no bandwidth: the rotation is then certified as a 0.074-approximate symmetry of the linear model (0.067 to 0.074 across filter windows, so not at $\delta = 0.05$), and exact rotation invariance is not excluded ($L = 0$). The OLS region, which ignores the dependence, and the Newey–West region, which allows serial correlation within each drifter's record, would both certify at $\delta = 0.05$ and exclude exact invariance (upper bounds 0.040 and 0.045, lower bounds 0.010 and 0.005). We report the Newey–West result as a sensitivity result only, because it depends on an estimated bandwidth and on asymptotics in the length of each record, whereas the cluster-robust region assumes nothing about the form or range of the dependence within a drifter. The dependence study above, in which Newey–West undercovered, used series of at most 100 points, much shorter than most drifter records here (median 1,486 pairs), so it is only indirect evidence. The verdict at $\delta = 0.05$ depends on this choice. Quadratic or cubic terms widen the cluster-robust bound to 0.11 and 0.21, and the construction of Proposition 4 applied over degrees 1 to 3 with cluster-robust regions gives 0.25 (the proposition is proved for the exact regions; with these regions the bound holds only asymptotically). The result rests on assumptions the data cannot check: a linear model class; velocities without measurement error (their median standard error is 0.014 $\mathrm{m\,s^{-1}}$ against a root-mean-square of 0.109 $\mathrm{m\,s^{-1}}$); errors uncorrelated with the velocity, although they contain the unmodelled wind forcing, which also drives the velocity (rotation-invariant forcing statistics would bias the fit without creating a defect; anisotropic forcing could create or hide one); independence between drifters, which share wind forcing within a region and season; and an asymptotic region with 31 clusters, five times the 6 coefficients of the linear class but only 2.6 and 1.6 times those of the quadratic and cubic classes. We report it as a case study within those assumptions.

3 Discussion

In practice the procedure changes what is reported. Fix the region ν where the symmetry is supposed to hold, the scales of the state variables, the candidate class and a tolerance δ on the relative defect. Fit a model class that is assumed to contain the truth, or certify simultaneously over a nested family whose largest member is (Proposition 4). Compute $U(\theta)$, certify what falls below δ, and report U itself rather than a binary verdict, because it is the smallest tolerance the data support and the honest analogue of the singular value that would otherwise be reported; report the extrapolation factor $\sqrt{\kappa\kappa'}$ of Proposition 6 so the reader can see how much of the

target domain the data actually visited. The reporting checklist in Methods lists every choice that gives the tolerance its meaning.

The limitations follow from Table 1. The model class is the binding assumption: validity requires the truth to lie in the fitted class, in the largest class of a nested family, or within an assumed distance of it. Paying for that distance can leave nothing to report: on a non-polynomial flow with an exact rotational symmetry the narrowest bound that accounted for model error was 0.27, against a tolerance of 0.05 (Supplementary Fig. S2), so in practice the route is a class or dictionary that contains the truth (Supplementary Table S6). A fitted model can be more symmetric than the system it models, and no analysis of that model alone can detect the difference. A lack-of-fit test that fails to reject does not establish correct specification: Proposition 3 quantifies the misspecification it misses, and in Section 2.8 misspecification missed with probability one half can produce true defects about 1.7 times the tolerance. Exact coverage of the coefficient region needs Gaussian homoskedastic errors; the sandwich regions that allow heteroskedastic or dependent errors are valid only asymptotically, and in our designs they undercovered (White's region 0.77 to 0.88 at $N = 400$ under every error distribution tried, the cluster-robust region 0.73 to 0.92 with 48 trajectories under autocorrelated errors, the Newey–West region worse still on trajectories of at most 100 points, where the certificate built on it covered in as few as 80% of replications), so dependent data need many independent trajectories (Supplementary Fig. S1, Supplementary Table S9). Noisy states are not covered: least squares then targets a pseudo-true field, which on the hardening Duffing oscillator at state noise $\sigma_x = 0.08$ is more symmetric than the system, so that the certificate concentrates on a spurious symmetry as data accumulate, while on the symmetric Hopf system state noise hides the symmetry instead; integral or weak formulations of model discovery [39], ensemble resampling [40] and explicit errors-in-variables methods [41] are the natural routes, and none is part of the guarantee here. When its coefficient region covers, the certificate is conservative: in the tightness study its interval for the plug-in directions contained the true defect in every replication at every nominal level from 0.5 to 0.99, and for a generator fixed in advance at an exact symmetry its coverage was at least 0.93 at nominal 0.80 and 0.985 at nominal 0.95. In the degree-3 class its certified tolerance stayed a nearly constant factor of about 8 above the Le Cam lower bound across N and σ; in the classes of degree 5 and 7, examined at one noise level, the median factor rose to 10.9 and 13.2, and the pointwise bound for a generator fixed in advance grew almost as much, so the gap is not mainly a cost of simultaneity; it reflects the model class and the looseness of the two-point bound (Section 2.9). We treat connected symmetry groups whose generators lie in a prescribed finite-dimensional class, excluding translations, for which the relative defect degenerates, and moderate problem sizes ($n \leq 6$, $Q \leq 210$, $P \leq 36$). Discrete symmetries, and symmetries visible only after a nonlinear change of coordinates [42, 43], are outside the framework, as they are outside the Lie-algebraic setting generally. Finally, the benchmarks are simulated deliberately, because measuring an error rate requires knowing the truth exactly, which a real system does not provide. Of the two real data sets we analyse, the lynx–hare records are too small to support any symmetry claim (Supplementary Note 2), and the drifter

velocities certify a physically expected rotation invariance to a stated tolerance, but only under assumptions the data cannot check (a model class, an observation model, errors uncorrelated with the velocity although they contain the unmodelled wind forcing, independence between drifters, and an asymptotic region with 31 clusters); neither replaces a setting in which symmetries are both real and known.

The work sits among several lines of research. The idea that continuous symmetries are a nullspace runs through work on point clouds [4], conserved quantities [7], trained networks [5, 6] and dynamic mode decomposition [44]; Otto et al. [8] unify these, extend them to nonlinear group actions on manifolds and to sections of vector bundles, and add convex penalties that promote symmetry during training. Their conclusion names our problem: it will be important "to study the perturbative effects of noisy data in algorithms to discover and promote symmetry, with the goal of understanding the effects of problem dimension, noise level, and amount of data." We answer with confidence statements rather than with a perturbation analysis, because a perturbation bound alone does not tell a user what to declare; the Weyl-inequality baseline in our experiments is the perturbation answer, and it is valid but never certifies. They also ask whether their framework can "find elements that are nearly symmetric, and to bound the degree of asymmetry"; Definition 1 is such a bound and SymCert supplies its confidence statement, in a setting where approximate symmetry is widely recognised in modelling [45, 46]. An alternative line searches for generators by nonlinear optimisation or adversarial training [42, 43, 47–49], and a related line learns invariances as hyperparameters through augmentation or marginal likelihood [50, 51]. These methods return a point estimate of a generator with no error control, and the question asked here, when a discovered generator may be declared and at what tolerance, applies to them too: whenever the Lie derivative is bilinear in the generator coordinates and the model coefficients the machinery transfers, provided the coefficient region is itself valid (for learned features, that requires features fixed independently of the data used for the region; see 'Learned features' in Section 2.3), and for a fully nonlinear model the ellipsoid must be replaced by a confidence region obtained some other way, with the logic of Proposition 1 unchanged. Once a symmetry is in hand it is enforced as hard constraints or convex penalties [8], as physics-informed losses [52, 53], as data augmentation [54], or inside equation discovery itself [55], and everything downstream inherits the reliability of the discovery step. On the statistical side, equivalence testing is standard wherever the interesting hypothesis is the null [11–13]; our contribution is not the idea but the observation that symmetry discovery is one of those settings, with a computable construction for this functional. Rank tests [9, 10, 56] attack the same nullspace-dimension question and place symmetry in the null. The selection of a generator is the issue addressed by post-selection and split-sample inference [20–23, 32]; here simultaneity over the whole candidate class dissolves it more cheaply than splitting, which we verify empirically, whereas selection of the model class needs the separate treatment of Proposition 4, in line with the general difficulty of inference after model selection [57]. Because a single event covers all generators at once, no multiplicity correction of the Benjamini–Hochberg kind [58] is needed. Finally, the shape of the problem (many moment conditions, a

low-dimensional parameter, and a test of whether they can all be satisfied) is that of a test of overidentifying restrictions [59], with the difference that here the restriction of interest is the one we would like to accept rather than reject.

Symmetry discovery is an instance of a pattern that recurs whenever a scientific claim takes the form "this quantity is zero": a rank, a conservation law, a conditional independence, a null effect. In each, an estimate is compared to a tolerance, and the tolerance is asked to carry statistical weight it cannot bear. The remedy is the same each time: state the tolerance in interpretable units, and require the data to *exclude* the alternative instead of merely failing to detect it. Whether that remedy is cheap depends on the functional; for the Lie derivative of a model that is linear in its coefficients it costs a sequence of trust-region subproblems, and its reach is set by how honestly the model class and the observation model can be specified.

4 Methods

4.1 Construction of the bilinear forms

Polynomial models.

Represent a polynomial vector field on $\mathbb{R}^n$ by an $n \times m$ array of coefficients over the monomials of total degree at most d. Multiplication and differentiation of monomials are index tables computed once, so the bracket (1) of two such fields is exact integer and rational linear algebra. Fixing the candidate basis $\zeta_1, \dots, \zeta_P$ and the model basis $\{x^a e_i\}$ and computing the bracket of every pair gives a three-index tensor $T \in \mathbb{R}^{J \times P \times Q}$, with J the number of coefficients of a bracket, such that $\mathrm{coef}([\xi_\theta, F_\beta]) = T \cdot \theta \cdot \beta$, and likewise $T^{(1)}, T^{(2)}$ for the two individual terms, with $T = T^{(1)} - T^{(2)}$. Multiplying the first index by $W^{1/2} \otimes g_\nu^{1/2}$, where g_ν is the monomial Gram matrix of ν and W the output metric, converts Euclidean norms of contracted vectors into the norms $\|\cdot\|_{\nu,W}$ of the corresponding fields. Then $L_\theta = T_w \cdot \theta$, and the columns of $A(\beta) = T_w \cdot \beta$ give $C(\beta) = A(\beta)^\top A(\beta)$. The matrix $M(\beta)$, defined by $M(\beta)\theta = M_\theta \beta$, stacks $T_w^{(1)} \cdot \beta$ and $T_w^{(2)} \cdot \beta$ and gives $D(\beta) = M(\beta)^\top M(\beta)$. The matrix g_ν is available in closed form for the uniform measure on a box (with equal or unequal half-widths), the Gaussian, and the uniform measure on a ball or annulus (the last two by separating radial and angular moments), so no quadrature error enters. For the *data* measure we substitute the empirical Gram matrix of the same monomial basis; the difference between the two is exactly what Proposition 6 measures. To evaluate the extrapolation factor, G_μ is symmetrised and treated as singular when its smallest eigenvalue is at most 10^{-10} times its largest, in which case $\sqrt{\kappa\kappa'}$ is reported as infinite; otherwise κ and κ' are computed with symmetric inverse square roots. No pseudoinverse or ridge is used.

Dictionary models.

For $F_\beta = \sum_{i,a} \beta_{i,a} \phi_a e_i$ with differentiable ϕ_a, the two terms of the bracket of ξ with $\phi_a e_i$ are $(\nabla \phi_a \cdot \xi)\, e_i$ and $\phi_a\, (D\xi)\, e_i$, both linear in β pointwise. Evaluating them at the nodes x_k of a quadrature rule with weights w_k for ν and stacking $\sqrt{w_k}\, W^{1/2}$ times their values gives tensors with the same role as $T_w^{(1)}, T_w^{(2)}$; their output index is

compressed by an orthonormal basis of the span of all contracted vectors, which leaves every norm unchanged. A uniform grid of p points per angle integrates trigonometric polynomials of degree below p exactly, and a Gauss–Legendre rule with p nodes integrates polynomials of degree up to $2p-1$ exactly. For the radial library we integrate in polar coordinates, Gauss–Legendre (40 nodes) in the radius with the area weight and a uniform grid (24 points) in the angle, and for the pendulum library a 32×32 Gauss–Legendre product rule. For the radial family and the pendulum the test suite checks that doubling both rules changes no defect by more than a relative 10^{-10}, and that Monte Carlo integration of the analytic bracket over the target measure (10^6 points) agrees with the quadrature defects to within three standard errors. With a monomial dictionary and an exact rule the construction reproduces the closed-form tensors to 10^{-10}, which the test suite also checks.

4.2 Certified numerics

All extrema are over the ball $\|u\| \le R$ that parameterises the ellipsoid. For $\max \|a + Bu\|$ we use the dual bound $\|a + Bu\|^2 \le a^\top a + \omega R^2 + c^\top (\omega I - B^\top B)^{-1} c$, valid for every multiplier $\omega > \lambda_{\max}(B^\top B)$ with $c = B^\top a$; for $\min \|a + Bu\|$ the bound $\|a + Bu\|^2 \ge a^\top a - c^\top (B^\top B + \omega I)^{-1} c - \omega R^2$, valid for every $\omega \ge 0$ with the matrix positive definite (and the least-squares residual when the unconstrained minimiser lies in the ball). For the indefinite problems of Proposition 2(ii) we use the dual function $\psi(\omega) = s_0 - g^\top (H + \omega I)^{-1} g - \omega R^2$ of $\min\, u^\top H u + 2 g^\top u + s_0$, valid and concave for every $\omega \ge 0$ with $H + \omega I$ positive definite (at the end point $\omega = -\lambda_{\min}(H) > 0$ it remains valid when g is orthogonal to the bottom eigenspace of H; otherwise that end point is approached as a limit). In each case the multiplier is located by a safeguarded Newton iteration on the derivative of the dual (the secular equation), in the reciprocal form of Moré and Sorensen, which is concave and nearly linear, with bisection as a fall-back [14, 15, 24]; whatever multiplier is returned, the value is a valid bound, and the minimiser of the subproblem (with a component along the bottom eigenvector in the hard case) is a feasible point. The sharp certificate uses this in Dinkelbach's iteration for fractional programmes [60]: starting from the plug-in defect, each step solves the indefinite subproblem at the current level t and moves t to the squared ratio at its minimiser, a feasible point, so the iterates increase to $U(\theta)^2$ from inside; the square root of the final value, inflated by a relative 10^{-7}, is accepted only when the dual bound certifies $\min_u t\|M_\theta \beta(u)\|^2 - \|L_\theta \beta(u)\|^2 > 0$ there, and bisection between the attained value and the separate bound takes over otherwise. A further relative margin of 10^{-9} covers rounding, and the analogous construction from above gives $L(\theta)$. A denominator whose lower bound is below 10^{-12} of its scale is treated as vanishing, and the generator as not certifiable. The separate bound uses $\sup_{\mathcal{B}} \|L_\theta \beta\|$ and $\inf_{\mathcal{B}} \|M_\theta \beta\|$ (and, when model error is admitted, the two denominator blocks separately, because the model-error terms perturb them by different amounts). One sharp certificate costs about 3 ms of CPU time for $Q = 20$ and 10 ms for $Q = 60$ on one core.

For subspaces, with V also denoting a matrix whose orthonormal columns span the subspace and θ ranging over its unit vectors, we bound $\sup_\theta \|A(\beta)\theta\| \le \sigma_{\max}(A(\hat\beta)V) + \sup_{\mathcal{B}} \|(A(\beta) - A(\hat\beta))V\|_F$ and

$\inf_\theta \|M(\beta)\theta\| \geq \sigma_{\min}(M(\hat\beta)V) - \sup_{\mathcal{B}} \|(M(\beta) - M(\hat\beta))V\|_F$, each supremum being the norm of a linear image of β and hence one more trust-region maximum. The refutation bound uses the same construction with $V = I$, the whole candidate class: $\min_\theta \rho_\nu \geq (\sigma_{\min}(A(\hat\beta)) - r_A)_+ / (\sigma_{\max}(M(\hat\beta)) + r_M)$ on $\{\beta^\star \in \mathcal{B}\}$, with $r_A = \sup_{\mathcal{B}} \|A(\beta) - A(\hat\beta)\|_F$ and $r_M = \sup_{\mathcal{B}} \|M(\beta) - M(\hat\beta)\|_F$. Generalised eigenproblems are solved on the range of $D(\hat\beta)$; directions in its null space (relative eigenvalue at most 10^{-12}) have an undefined defect and are never declared or certified.

Pointwise bound.

For a generator fixed in advance, or chosen on one half of the data and bounded with the fit to the other half, the error probability is split into 0.98α for the numerator, 0.01α for the denominator and 0.01α for the noise scale: $\hat\Sigma$ is multiplied by the upper confidence factor $\mathrm{dof}/\chi^2_{\mathrm{dof};\,0.01\alpha}$ (1 for the asymptotic regions), and $\|L_\theta\hat\beta\|$ and $\|M_\theta\hat\beta\|$ are each moved up and down by the Gaussian-concentration width $(\operatorname{tr}\Gamma)^{1/2} + (2\|\Gamma\|_{\mathrm{op}}\log(1/\alpha'))^{1/2}$ at their share α', with Γ the inflated covariance of $L_\theta\hat\beta$ or $M_\theta\hat\beta$. The bounds on the defect are the ratios of these bounds. When model error is admitted, the denominator's share is spent on its two blocks, 0.005α each.

Closure diagnostic.

For a certified plug-in subspace V_r with $r \geq 2$ we take a basis $\xi_1, \ldots, \xi_r$ that is orthonormal in the $\|\cdot\|_{\nu,W}$ geometry of generator fields and compute each commutator $[\xi_a, \xi_b] = D\xi_b\,\xi_a - D\xi_a\,\xi_b$ exactly. For affine generators it lies in the candidate class, which is closed under the bracket. The reported closure defect is the largest over pairs of

$$c(a,b) = \frac{\|P_\perp[\xi_a,\xi_b]\|_{\nu,W}}{\left(\|D\xi_b\,\xi_a\|^2_{\nu,W} + \|D\xi_a\,\xi_b\|^2_{\nu,W}\right)^{1/2}},$$

with $P_\perp$ the orthogonal projection, in the same geometry, onto the complement of V_r in the candidate class. For a class closed under the bracket, as here, $c = 0$ for every pair exactly when V_r is a Lie subalgebra, and $c \leq \sqrt{2}$. A commutator whose norm exceeds 5% of the same denominator is itself a candidate generator, and its defect is bounded by the sharp certificate from the same region at the same level; the commutators count as certified when every such bound is at most δ. Smaller commutators, as in an abelian algebra, are reported as negligible. For a candidate class that is not closed under the bracket (quadratic generators, for example), a commutator need not be a candidate generator and then cannot be certified, and c measures only its part inside the class: the part outside it must be reported alongside c, and $c = 0$ alone does not establish closure.

4.3 Confidence regions

Unless stated otherwise the region is the exact least-squares ellipsoid of Proposition 2(iii) ("OLS" below). The sandwich regions use $\hat\Sigma = (I_n \otimes (\Phi^\top\Phi)^{-1})\,\Omega\,(I_n \otimes (\Phi^\top\Phi)^{-1})$ with scores $z_t = r_t \otimes \phi(x_t)$, where $r_t \in \mathbb{R}^n$ is

the least-squares residual of row t. White's estimator (HC0) takes $\Omega = \sum_t z_t z_t^\top$ and the radius $R^2 = \chi^2_{Q;\,1-\alpha}$. The cluster-robust estimator is the jackknife (CR3) form recommended when clusters are few [61]: with n_c trajectories, $\Omega = \frac{n_c-1}{n_c}\sum_g s_g s_g^\top$, where $s_g = \Phi_g^\top (I - H_{gg})^{-1} r_g$ collects the residuals of trajectory g inflated by its leverage block $H_{gg} = \Phi_g(\Phi^\top\Phi)^{-1}\Phi_g^\top$; it uses the small-sample radius $R^2 = Q\,\frac{n_c-1}{n_c-Q}\,\mathrm{F}_{Q,\,n_c-Q;\,1-\alpha}$, the Hotelling form, which tends to the chi-square quantile as n_c grows. The Newey–West estimator adds, within each trajectory, the products of scores at lags $j = 1, \ldots, \ell$ with Bartlett weights $1 - j/(\ell+1)$, with ℓ chosen by Andrews' AR(1) plug-in rule from the pooled lag-one autocorrelation of the residuals, and uses the chi-square radius [30, 31]. Directions outside the range of $\hat\Sigma$ (relative eigenvalue at most 10^{-12}), such as those missed by a cluster-robust estimate with fewer trajectories than coefficients, and the coefficient directions that a rank-deficient design cannot see are treated as unidentified, and any certificate that depends on them is infinite. Region coverage is checked by evaluating $(\hat\beta - \beta^\star)^\top \hat\Sigma^{-1} (\hat\beta - \beta^\star) \le R^2$ against the known $\beta^\star$. The estimators used are: OLS in every experiment except the following; White and OLS for the error-distribution study; White and cluster-robust for the trajectory study; OLS, White, cluster-robust and Newey–West for the serial-dependence study and for the drifter case study; White for the lynx–hare case study, where with 19 rows it carries no asymptotic justification.

4.4 Model-class procedures

The lack-of-fit test is the F-test of the smaller class against the larger on the same rows, with residual sums of squares from both fits, $n(m_{\mathcal{H}} - m)$ and $n(N - m_{\mathcal{H}})$ degrees of freedom, and level 0.05. Its denominator is the residual mean square of the largest class, and the certificate uses the residual variance of the class in which it is computed, so the two share one estimate when the gate selects the largest class and use different estimates from the same rows otherwise; under *Split* they use disjoint halves of the rows, and *Largest* and *Simultaneous* run no test. The *gate* chooses the smallest class whose test against the largest does not reject and certifies there; no error budget is allocated to the test, because the gate carries no guarantee. *Split* applies the gate to the first half of the rows and certifies on the second. *Largest* certifies in the largest class at level α. *Simultaneous* implements Proposition 4 with α/K per class; the omitted part of class k is computed from the least-squares projection of the largest class on class k over the design, and its $\|\cdot\|_{\nu,W}$ norm is maximised over the largest-class region. The invisible defect of Proposition 3 is evaluated with $\mathcal{H}$ the largest class, μ_N replaced by ν (the designs are uniform on the target box), $\gamma = 0.5$ and the denominator of (8) evaluated at F_{β° (the degree-3 Hopf field), that is, to first order in h. The resulting values describe one omitted part, $h^\star$, and are not upper bounds; evaluating the denominator exactly at $F_{\beta^\circ} \pm h^\star$ gives 0.039–0.040 (degree 5) and 0.085–0.089 (degree 7) at $N = 3200$, depending on the sign of $h^\star$.

4.5 Baselines

Fixed threshold: declare the span of all generalised eigenvectors of $(C(\hat\beta), D(\hat\beta))$ whose plug-in defect is at most τ; sweeping τ traces the whole family, including

$\tau = \delta$. *Eigengap*: declare the r maximising $\rho_{(r+1)}/\rho_{(r)}$ over $r \geq 1$; this rule cannot express "no symmetry", which is one reason its error rate is high. *Weyl perturbation certificate*: bound $\lambda_k(C(\beta^\star)) \leq \lambda_k(C(\hat{\beta})) + \|C(\beta^\star) - C(\hat{\beta})\|$ with the perturbation bounded by $2\|A(\hat{\beta})\|\,\|\Delta A\|_F + \|\Delta A\|_F^2$, $\|\Delta A\|_F$ maximised over the same region, and divide by a lower bound on $\lambda_{\min}(D)$; this is the natural singular-value perturbation answer [62–64], valid and loose because it spends one global perturbation budget on every direction. *Rank test*: for each r, test H_0: the r smallest plug-in directions span an exact nullspace, using the squared norm of the restricted bracket referred to a Welch–Satterthwaite approximation of its null distribution, and report the largest r not rejected, a Wald-type rank test in the spirit of Robin and Smith [9] and Kleibergen and Paap [10]. *Split significance test*: select the smallest-defect direction on one half of the data, test H_0: exact symmetry on the other half by the same statistic, and declare a symmetry when the test fails to reject. SymCert *with sample splitting*: fit the first and the second half of the rows separately, take the plug-in generalised eigenvectors of the first-half fit in order of plug-in defect, certify the span of the first $r = 1, 2, \ldots$ of them with the simultaneous certificate computed from the second-half region (the sharp certificate for $r = 1$, the subspace bound otherwise), and declare the largest r before the first span whose bound exceeds δ. In the dimension study only the first direction is bounded, with the pointwise bound. *Parametric bootstrap*: draw $\beta^* \sim N(\hat{\beta}, \hat{\Sigma})$ 400 times and report the α and $1-\alpha$ quantiles of $\rho_\nu(\xi_\theta; F_{\beta^*})$ [35].

4.6 Benchmarks, metrics and Monte Carlo uncertainty

Ten polynomial systems on $\mathbb{R}^2$ and $\mathbb{R}^3$ serve as the main benchmarks (all defined in the released code). Four have a nonzero symmetry algebra inside the affine class, of dimension $r^\star$: the Hopf normal form $\dot{x} = x - y - x(x^2+y^2)$, $\dot{y} = x + y - y(x^2+y^2)$ ($r^\star = 1$), the planar rotation $\dot{x} = -y$, $\dot{y} = x$ ($r^\star = 2$), the Euler equations of a symmetric top with moments of inertia $(1, 1, 2)$ ($r^\star = 2$) and the radial flow onto the sphere $\dot{x} = x(1 - |x|^2)$ on $\mathbb{R}^3$ ($r^\star = 3$, $\mathfrak{so}(3)$). Six have none: Van der Pol ($\dot{x} = y$, $\dot{y} = -x + y - x^2 y$), Duffing ($\dot{x} = y$, $\dot{y} = x - x^3 - y/5$), Lotka–Volterra ($\dot{x} = x - xy$, $\dot{y} = xy - y$), Sel'kov ($\dot{x} = 1/10 - x + y/2 + x^2 y$, $\dot{y} = -y/2 - x^2 y$), the Euler equations of an asymmetric rigid body with moments $(1, 2, 3)$, and Lorenz with parameters 10, 28 and 8/3. Each is fitted in the polynomial class of its own degree (1 for the rotation, 2 for the rigid bodies, Lotka–Volterra and Lorenz, 3 otherwise) with affine candidate generators. The target measure ν is uniform on the box $[-s, s]^n$, with $s = 2$ for Duffing and Sel'kov, $s = 20$ for Lorenz and $s = 1$ otherwise (except in the selected-generator study of Supplementary Fig. S5e, where all four systems are on $[-1, 1]^2$, and in the trajectory study of Supplementary Table S3, where Van der Pol is on $[-3, 3]^2$ and Hopf on $[-1.5, 1.5]^2$); the states are drawn from ν and the errors are Gaussian. The calibration grid crosses $N \in \{100, 200, 400, 800, 1600, 3200\}$ with noise levels $\{0.01, 0.03, 0.1, 0.3\}$. The broken-Hopf family is $F_\varepsilon = F_{\text{Hopf}} + \varepsilon x^2 \partial_x$ (the term εx^2 enters $\dot{x}$ only); the rotation defect grows continuously from zero with ε. Every system has rational coefficients, so its exact symmetry algebra is the exact rational nullspace of the bracket map, computed with symbolic arithmetic, and the true defect of any generator follows in closed form from monomial moments; the algebra dimensions are cross-checked against an independent characterisation (for linear systems, the

dimension of the commutant of the coefficient matrix). For the dictionary families the true algebra is the numerical nullspace of the exact bracket map, with a gap of more than ten orders of magnitude between its zero and nonzero singular values.

The truncation study fits Van der Pol, Duffing, Hopf and Sel'kov on their boxes in the polynomial classes of degree 1, 2 and 3 (the correct class, as a control), with $N \in \{400, 1600\}$, noise level 0.02 and 300 replications per setting; the lack-of-fit test compares each fit with the cubic class. The sparsity study fits the broken-Hopf members with $\varepsilon \in \{0, 0.02, 0.05, 0.1, 0.2, 0.4\}$ on $[-1, 1]^2$ in the cubic class by sequentially thresholded least squares (STLSQ): a least-squares fit followed by ten rounds in which every monomial coefficient of magnitude below the threshold is set to zero and each component is refitted on the remaining monomials, with thresholds 0 (plain least squares), 0.05, 0.1 and 0.25, $N = 1000$, noise level 0.05 and 300 replications per setting; the threshold rule is applied to the STLSQ fit and SymCert to the unpruned least-squares fit of the same data. The noise study near the tolerance uses broken-Hopf members with ε tuned so that the smallest true defect in the affine class is 1.2, 1.5, 2 or 4 times δ, and the hardening Duffing oscillator $\dot{x} = y$, $\dot{y} = -x - 0.3x^3 - 0.2y$ (smallest true defect 0.102), all on $[-1, 1]^2$ in the cubic class, with $N \in \{100, 400, 1600\}$, noise levels $\{0.05, 0.1, 0.2, 0.3\}$ and 300 replications in each of the 60 settings; as no generator is within tolerance, every declaration by the threshold rule at δ and every certification is false.

False certification is the event that the declared subspace contains a generator whose true defect exceeds δ; because the declared object is a subspace, this is evaluated as a generalised eigenvalue of the true forms restricted to it, not by sampling directions. *Detection* is the event that the declared dimension is at least the true one, and *recovery* the event that, in addition, the declared subspace contains the true algebra up to a largest principal angle of 10° in the $L^2(\nu)$ geometry of generator fields. Unless stated otherwise $\delta = \alpha = 0.05$.

Where an interval is given for a rate estimated from k events in T replications, it is the exact Clopper–Pearson interval [65]; when no event occurs, the one-sided 95% upper bound is $1 - 0.05^{1/T}$ (0.010 for $T = 300$). For a rate given without an interval, the 95% Clopper–Pearson interval extends at most 0.058 from the estimate for $T \geq 300$, 0.072 for $T \geq 200$ and 0.116 for $T \geq 80$, the smallest number of replications used. Simultaneous statements over S settings use the Bonferroni level $0.05/S$ per setting. These intervals quantify Monte Carlo error only. The validity of the certificate rests on Proposition 1; the simulations check that the implementation and the assumptions behave as the theorem says, and probe what happens when the assumptions fail.

4.7 Ocean drifter data

Hourly positions and velocities are from the Global Drifter Program product of Elipot et al. [37], version 2.01 [38], obtained from the NOAA AOML ERDDAP server on 26 October 2026 for 30–45°N, 180–140°W and 1 June to 30 September of 2016–2019. We kept rows recorded before the loss of the drogue and with interpolation gaps of at most six hours, split each drifter's record into contiguous hourly segments of at least 150 hours, removed a centred running mean (49 hours

unless stated) to isolate the near-inertial band, and took central differences for the acceleration, dividing by the local Coriolis parameter so that time is measured in units of $1/f$; velocities are expressed in units of their root-mean-square. Every step acts identically on both velocity components and commutes with rotations, so the processing introduces no preferred direction and cannot remove a rotation invariance that the data have. Removing the running mean does discard the sub-inertial flow (mean, geostrophic and Ekman currents), which need not be rotation invariant, so the certificate concerns the dynamics of the high-passed, near-inertial velocities, not of the full flow. The model classes are polynomials of degree 1 to 3 in the velocity with affine candidate generators, the target measure is the standard Gaussian in the normalised velocity (extrapolation factor 1.02 for the linear class), and the regions are OLS, White, cluster-robust over drifters (31 clusters) and Newey–West within drifters, applied to each drifter's segments concatenated in time order. The processed subset is distributed with the code.

4.8 Reporting checklist

A certificate is interpretable only together with the choices that give the tolerance its meaning. We recommend reporting, in this order:

1. the target measure ν (the region the claim is about) and the output metric W, or the characteristic scales used to nondimensionalise the state;
2. the candidate class of generators;
3. the model class or dictionary, how it was chosen (fixed in advance, by Proposition 4, or by a heuristic), and any model-error bound with its justification;
4. the observation model and the confidence region used (exact, White, cluster-robust, Newey–West), with the regime in which it is valid;
5. the extrapolation factor $\sqrt{\kappa\kappa'}$ of the observed states;
6. $U(\theta)$ for every generator reported, the tolerance δ and whether it was fixed before the analysis, the dimension of the certified plug-in subspace, and, if it is at least two, its closure diagnostic;
7. the lower bound $\min_\theta L(\theta)$ when the claim is that no symmetry exists.

4.9 Proofs

Lemma 1.

Write $u = DF\,\xi$ and $v = D\xi\,F$ as elements of $L^2(\nu)$ with the inner product induced by W. Then $\rho^2 = \|u - v\|^2/(\|u\|^2 + \|v\|^2)$. The inequality $\|u - v\|^2 \le 2(\|u\|^2 + \|v\|^2)$ gives $\rho \le \sqrt{2}$; $\rho = 0$ iff $u = v$, i.e. $[\xi, F] = 0$ ν-almost everywhere, which for continuous fields means on the support of ν. When the support is the closure of an open set, the bracket vanishes on its interior, which is equivalent to the flow of ξ mapping solutions to solutions there [66, Ch. 9]; for polynomial or other analytic fields a bracket that vanishes on an open set vanishes identically. On a lower-dimensional support no such conclusion follows. Under $\xi \mapsto c\xi$ both u and v scale by c, under $F \mapsto c'F$ by c', so the ratio is invariant, and hence also under any invertible linear reparameterisation of the candidate class, because ρ depends on θ only through ξ_θ. For $x = Az + b$ the

pulled-back fields are $\tilde{F}(z) = A^{-1}F(Az+b)$ and $\tilde{\xi}(z) = A^{-1}\xi(Az+b)$, with $D\tilde{F}(z) = A^{-1}DF(Az+b)A$ and likewise for ξ, so $\tilde{u}(z) = A^{-1}u(Az+b)$ and $\tilde{v}(z) = A^{-1}v(Az+b)$. With $\tilde{\nu}$ the transported measure and $\tilde{W} = A^\top W A$, $\int \tilde{u}^\top \tilde{W}\tilde{u}\, d\tilde{\nu} = \int u^\top W u\, d\nu$ and likewise for v and $u-v$, so the ratio is unchanged. With $\tilde{W} = W = I$ this requires $A^\top A \propto I$, i.e. a similarity; for any other A the scalings of the components differ and the ratio changes in general (Supplementary Table S1). □

Lemma 2.

DF_β and F_β are linear in β, and ξ_θ and $D\xi_\theta$ are linear in θ, so each of $DF_\beta\,\xi_\theta$ and $D\xi_\theta F_\beta$ is bilinear in (θ, β) pointwise. Their $\|\cdot\|_{\nu,W}$ norms are therefore quadratic forms in β for fixed θ and in θ for fixed β, and the symmetric square roots of the Gram matrices give the matrices of the lemma. $\rho^2 = \theta^\top C\theta / \theta^\top D\theta$; if $D\theta = 0$ then $M_\theta\beta = 0$, both terms vanish, $L_\theta\beta = 0$ and the defect is undefined. On the range of D the minimum of the Rayleigh quotient is the smallest generalised eigenvalue. □

Proposition 1.

On the event $E = \{\beta^\star \in \mathcal{B}\}$ we have $L(\theta) \le \rho_\nu(\xi_\theta; F_{\beta^\star}) \le U(\theta)$ for every θ with a defined defect, by definition of the infimum and supremum. Since $\mathbb{P}(E) \ge 1-\alpha$ and E depends neither on θ nor on δ, the statement holds simultaneously. For (6): on E, any θ and δ with $U(\theta) \le \delta$ have $\rho_\nu(\xi_\theta; F_{\beta^\star}) \le \delta$, whatever rule produced them, so the bad event is contained in E^c. □

Proposition 2.

(i) With $\beta = \hat{\beta} + \hat{\Sigma}^{1/2}u$ each norm is $\|a + Bu\|$ with $a = L_\theta\hat{\beta}$, $B = L_\theta\hat{\Sigma}^{1/2}$ (or the same with M_θ), and its extrema over the ball are trust-region subproblems; dividing the supremum of the numerator by the infimum of the denominator bounds the supremum of the ratio. (ii) The denominator being positive on the ball, $\sup_u \|L_\theta\beta(u)\|^2 / \|M_\theta\beta(u)\|^2 \le t$ iff $t\|M_\theta\beta(u)\|^2 - \|L_\theta\beta(u)\|^2 \ge 0$ for all u in the ball, iff its minimum over the ball is non-negative; the minimum of a quadratic over a ball has no duality gap, even when the quadratic is indefinite [15, 24], so the dual bound decides the sign up to the bisection tolerance. The lower bound $L(\theta)$ is obtained in the same way from the sign of the minimum of $\|L_\theta\beta(u)\|^2 - t\|M_\theta\beta(u)\|^2$. (iii) In model (3) with a fixed design Φ of full column rank, $\hat{\beta} - \beta^\star \sim N(0, \sigma^2\Sigma_0)$ with $\Sigma_0 = I_n \otimes (\Phi^\top\Phi)^{-1}$ and $\mathrm{RSS}/\sigma^2 \sim \chi^2_{\mathrm{dof}}$ independent of $\hat{\beta}$, so $\|\hat{\Sigma}^{-1/2}(\hat{\beta} - \beta^\star)\|^2/Q$, with $\hat{\Sigma} = \hat{\sigma}^2\Sigma_0$, is exactly $\mathrm{F}_{Q,\mathrm{dof}}$ distributed. □

Proposition 3.

The F-statistic compares the drop in residual sum of squares from the fitted class to $\mathcal{H}$ with the residual mean square of $\mathcal{H}$; with Gaussian errors and the truth in $\mathcal{H}$ its numerator is a noncentral chi-square with noncentrality equal to the squared norm, summed over rows and components and divided by σ^2, of the part of F in $\mathcal{H}$ orthogonal to the fitted class over the design, which is $N\|h\|^2_{\mu_N}/\sigma^2$. The power is increasing in λ, which gives the first claim. Because F_{β° is the least-squares projection over the design, h lies in $\mathcal{H}_\perp$, so $\|[\xi, h]\|_{\nu,W} \le G(\xi)\|h\|_{\mu_N}$ by (7), and by

the triangle inequality the numerator $\|[\xi, F_{\beta^\circ} + h]\|_{\nu,W}$ differs from $\|[\xi, F_{\beta^\circ}]\|_{\nu,W}$ by at most this amount. Let $h^\star$ attain the maximum in (7) and scale it to $\|h^\star\|_{\mu_N} = \sigma(\lambda_\gamma/N)^{1/2}$; then $\|[\xi, h^\star]\|_{\nu,W} = G(\xi)\sigma(\lambda_\gamma/N)^{1/2}$, and the truth $F_{\beta^\circ} + h^\star$ has the same projection F_{β° and noncentrality λ_γ. If $[\xi, F_{\beta^\circ}] = 0$, the numerator of the defect of the truth is exactly $\|[\xi, h^\star]\|_{\nu,W}$, which gives (8). □

Proposition 4.

Work in model (3) and fix the design. In class $k < K$ the least-squares estimate satisfies $\hat\beta_k - \beta^\circ_k \sim N(0, \sigma^2\Sigma_{0,k})$, where β°_k are the coefficients of the least-squares projection of F on $\mathcal{F}_k$ over the design; the residual sum of squares is σ^2 times a noncentral chi-square (the noncentrality coming from the omitted part), independent of $\hat\beta_k$ and stochastically larger than the central one. The F-statistic of Proposition 2(iii) is therefore stochastically smaller than $\mathrm{F}_{Q_k,\mathrm{dof}_k}$, and $\mathcal{B}_k$ covers β°_k with probability at least $1 - \alpha/K$; $\mathcal{B}_K$ covers $\beta^\star$ with probability $1 - \alpha/K$. On the intersection of these K events, which has probability at least $1 - \alpha$, $F = F_{\beta^\circ_k} + h_k(\beta^\star)$ with $\|h_k(\beta^\star)\|_{\nu,W} \le \bar\eta_k$ for every k, and the L^2 model-error bound in class k gives $\rho_\nu(\xi_\theta; F) \le U_k(\theta)$. □

Proposition 5.

With L_ξ, M_ξ and $\Delta^\star$ as in the statement, set $\beta^t = \beta^0 + t\Delta^\star$. Since $L_\xi\beta^0 = 0$, $\|L_\xi\beta^t\| = t\, s_{\max}$, and $\|M_\xi\beta^t\| \le \|M_\xi\beta^0\| + t\|M_\xi\Delta^\star\|$, so $\rho_\nu(\xi; F_{\beta^t}) \ge t s_{\max}/(\|M_\xi\beta^0\| + t\|M_\xi\Delta^\star\|)$, which exceeds δ for every $t > t_\delta = \delta\|M_\xi\beta^0\|/(s_{\max} - \delta\|M_\xi\Delta^\star\|)$. The Kullback–Leibler divergence between the two data distributions is $\frac{Nt^2}{2\sigma^2}\|F_{\Delta^\star}\|^2_{\mu_N} = Nt^2/(2\sigma^2)$, so by Pinsker's inequality the total variation distance is at most $t\sqrt{N}/(2\sigma)$. For any rule, $\mathbb{P}_{\beta^0}(\varphi = 1) \le \mathbb{P}_{\beta^t}(\varphi = 1) + t\sqrt{N}/(2\sigma) \le \alpha + t\sqrt{N}/(2\sigma)$ for every $t > t_\delta$; letting $t \downarrow t_\delta$ gives (9), with no remainder term. This is the two-point method [67, Ch. 2]. □

Proposition 6.

Both $\|q\|^2_\nu$ and $\|q\|^2_\mu$ are quadratic forms in the coefficient vector c of $q \in \mathcal{R}$, equal to $c^\top G_\nu c$ and $c^\top G_\mu c$; their ratio is a generalised Rayleigh quotient whose maximum is $\lambda_{\max}(G_\mu^{-1/2} G_\nu G_\mu^{-1/2})$, attained at the corresponding eigenvector. The defect statement follows by applying the bound $\|q\|^2_\nu \le \kappa\|q\|^2_\mu$ to the numerator, since the bracket lies in $\mathcal{R}$, and the reverse bound $\|q\|^2_\mu \le \kappa'\|q\|^2_\nu$ to each of the two terms of the denominator, both of which lie in $\mathcal{R}$. If G_μ is singular with null vector $c \ne 0$ then $\|q_c\|_\mu = 0$ while $\|q_c\|_\nu > 0$, so $\kappa = \infty$; conversely $\kappa = \infty$ forces G_μ to be singular. □

4.10 Software, reproducibility and compute

The implementation (Python, NumPy, SciPy and SymPy) and all experiment and figure scripts are released together with the raw result files (Code availability). All experiments are deterministic given the seeds recorded in the code. Every table of numerical results in the manuscript and the Supplementary Information is written by a script directly from the result files, and the numerical results quoted in the running text are asserted against those files by a checking script, which fails if any

has drifted. A test suite verifies the bracket against finite differences, the closed-form moments against Monte Carlo integration, the defect against Monte Carlo evaluation of its defining integrals, the exact symmetry algebra of linear systems against the dimension of the commutant, the certified trust-region bounds and the sharp ratio bounds against brute-force search over the ball, the invariance of the defect under changes of units with a transported metric, the dictionary construction against the closed-form one, the coverage of the coefficient region and of the certificate against their nominal levels, and the closure diagnostic on exact Lie algebras. One complete pass of the experiments costs about 3.6 CPU-hours as recorded by the experiment runner. A large language model (Claude, Anthropic) was used to assist with writing the code, running and analysing the experiments and drafting the text; the author takes responsibility for the content.

Data availability

All data analysed in this study are either generated by the released simulation code or taken from two public sources of real measurements. The Hudson's Bay Company lynx and hare pelt counts for 1900–1920 used in the lynx–hare case study (Supplementary Note 2) are the values distributed with the Stan example models (`knitr/lotka-volterra/hudson-bay-lynx-hare.csv`, https://github.com/stan-dev/example-models) and are reproduced verbatim in `experiments/exp12_realdata.py`. The ocean drifter velocities used in the drifter case study are from the NOAA Global Drifter Program hourly product, version 2.01 (https://doi.org/10.25921/x46c-3620, licence CC BY 4.0), accessed through https://erddap.aoml.noaa.gov/gdp/erddap/index.html; the processed subset used here is distributed with the code (`data/`) together with the script that regenerates it. All raw result files are in the `results/` directory of https://github.com/Kendiukhov/certified-symmetry-discovery; the version used for this article is tagged `v2.0.0`.

Code availability

The implementation, the experiment scripts, the figure and table generators, the number-checking script and the test suite are released under the MIT licence at https://github.com/Kendiukhov/certified-symmetry-discovery (tag `v2.0.0`).

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Author contributions

I.K. is the sole author and is responsible for the conception, the theory, the implementation, the experiments and the manuscript.

Additional information

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